

Nietzsche

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1. Some Ideas to Cover

1.1. Death of God

There is always a common misunderstanding of this phrase, seeing it as blasphemy and disloyalty to religion. But after this statement, Nietzsche continues: "We have killed him." This is meant to express that with the rise of science, reason, and modern civilization, religion is no longer sufficient to explain the world or satisfy people's spiritual needs. This is a diagnosis of the cultural condition of his time, not an attack on or denial of religion. Nietzsche's conclusion is that old values no longer apply; we need to seek new values on our own.¹

1.2. Will to Power

Personally, I find the English term more straightforward. "Will" refers to wanting to do something, and "to do" gives us a sense of drive and desire to act. Will to Power refers to the most fundamental life force. This is a point I particularly like. The passionate vitality of life drives people to expand, create, and do things bigger and better. This is one of the most praiseworthy qualities of life! The statues of ancient Greece display this most primal life force. I myself also really enjoy running; although my talent is modest, when I run long distances on the track, it's very clear that this state is very healthy and joyful! I recall my high school teacher once saying: "There's nothing a ten-kilometer run can't solve."

Nietzsche's core idea is that **the essence of life is not the will to survive, but self-overcoming, the expansion of power, and self-creation — the fundamental drive by which humanity establishes values after the death of God.**

1.3. Problem of Socrates

The divergence between Nietzsche and Socrates lies in:

Attitudes toward Reason

Socrates: Reason is the supreme judge. He proposed that "the unexamined life is not worth living," believing that through logical debate and conceptual definition, one can find universal truth and thereby guide correct action. Reason is the sole path to happiness and goodness.

Nietzsche: Reason is merely a tool of instinct and the will to power, a surface phenomenon. He criticized Socrates for elevating reason to the point of suppressing instinct, seeing this as a "tyranny of reason." True wisdom comes from the body, intuition, and creativity, not from cold logic.

The Foundations of Morality Socrates: Morality is a cognitive issue. No one willingly does evil; wrongdoing is merely the result of ignorance. Once a person truly knows the "good," they will necessarily do it. Therefore, the core of moral education is knowledge.

Nietzsche: Morality is an expression of the will to power. So-called "good" is nothing more than the value by which the strong affirm themselves (master morality), while the Socratic logic of "reason = virtue" actually weakens noble instincts and paves the way for slave morality (compassion, equality, asceticism). He accused Socrates of being "a populist who subverted aristocratic values."

Attitudes toward Life and Tragedy Socrates: Advocate of clarity, light, and knowable order. In the *Republic*, he opposed the irrational passions and forces of fate in Homeric epics and Greek tragedy, believing they disturb the soul.

Nietzsche: In *The Birth of Tragedy*, Nietzsche directly criticized Socrates as the culprit behind the "death of tragedy." He argued that Socratic optimistic rationalism (the belief that everything can be understood and corrected by logic) killed

¹Gay Science, 125.

the “Dionysian spirit” that arose from the depths of life in Greek tragedy — the instinctive power that unites chaos, suffering, ecstasy, and creation. The true will to life precisely requires embracing suffering and uncertainty.

Attitudes toward Death Socrates: Faced death calmly. In the *Phaedo*, he viewed death as the liberation of the soul from the prison of the body, entering the realm of pure reason. Philosophy is “practice for death.”

Nietzsche: Vigorously affirmed life, including its pain and death. He proposed “dying at the right time” and opposed all fantasies of an afterlife or otherworld. He said: “Socrates wanted to die — it was not Athens, but he himself who forced the hemlock upon himself.” He believed Socrates was fundamentally weary of life and used rational argumentation to find an excuse for death.

1.4. Apollonian vs Dionysian

The Apollonian and Dionysian are two opposing yet complementary artistic-spiritual symbols proposed by Nietzsche in *The Birth of Tragedy*. The Apollonian is the god of the sun, representing order, reason, clarity, and form; the Dionysian is the god of wine, representing chaos, passion, irrationality, and the primal force of life. Nietzsche believed that **the greatness of Greek tragedy lies in its successful fusion of these two spirits: through Apollonian structure and form, it expresses Dionysian vitality and suffering, thereby creating a profound artistic experience**. This is also reflected in the Problem of Socrates.

1.5. Übermensch (Overman / Superman)

The Übermensch is one who can create values from within themselves, affirm life in its entirety (including pain and destruction), and is not bound by old morality. Like Zarathustra, they are a “bridge to the future” spanning the abyss of nihilism. The Last Man settles for comfort, equality, and a risk-free life, seeking only mild, small happiness; the Übermensch pursues danger, greatness, and self-overcoming. The Übermensch is an ideal that humanity should strive toward, not a racial or political label.

1.6. Eternal Recurrence

This is because Nietzsche believed that the components of the world are finite, and therefore, after a sufficiently long time, the same events will recur in cycles. Your present and past life will repeat infinitely many times — every bit of pain and joy will be experienced again. Would you curse or cheer?

Ethical significance: the final blow against all “otherworldly” hopes. If life is eternally repeating and has no ultimate purpose, then the only meaningful attitude is to say “yes” to every moment, to love one’s fate (*amor fati*). This is the standard for testing whether a person truly affirms life.

2. Assigned Reading Analysis

2.1. Reading I: *The Gay Science* (1882/87) §125, §341–344

2.1.1. §125: The Madman and the Death of God

This is one of Nietzsche’s most famous passages. A madman runs into the marketplace in broad daylight carrying a lantern, shouting “I seek God!” — only to be mocked by the atheists present. The madman then declares: “God is dead. God remains dead. And we have killed him.”

Core analysis:

- First, this statement is not blasphemy against religion, but a **diagnosis of the condition of modern civilization**. The rise of science, reason, and modernity has made the Christian God “no longer believable” — religion can no longer provide a convincing framework of meaning for the world.
- “We have killed him” means: this is not an external event descending upon us, but an act of humanity itself — we have, through enlightenment, science, and modern civilization, destroyed the foundations of religious faith with our own hands.

- The madman forces his audience to face the consequences with a series of pressing questions: “Whither is it moving now? Whither are we moving? Away from all suns? Are we not plunging continually? Backward, sideward, forward, in all directions? Is there still any up or down? Are we not straying as through an infinite nothing?”
- Nietzsche’s conclusion: **Old values are no longer applicable; we need to seek new values on our own.** The death of God is both an unprecedented crisis and an unprecedented freedom.

2.1.2. §341: The Greatest Weight — The Thought Experiment of Eternal Recurrence

“What, if some day or night a demon were to steal after you into your loneliest loneliness and say to you: “This life as you now live it and have lived it, you will have to live once more and innumerable times more; and there will be nothing new in it, but every pain and every joy and every thought and sigh and everything unutterably small or great in your life will have to return to you, all in the same succession and sequence...” Would you not throw yourself down and gnash your teeth and curse the demon who spoke thus? Or have you once experienced a tremendous moment when you would have answered him: “You are a god and never have I heard anything more divine.””

This thought experiment is Nietzsche’s ultimate test of “life affirmation.” Suppose your life will repeat forever in exactly the same way — every pain, every regret, every small shame will be relived infinitely — how do you respond? If you curse the one who proposed this, if you curse fate — then deep down you **do not truly affirm life**. But if you can say “yes” to this fate, this is **amor fati** (love of fate): an unreserved affirmation of every moment of existence.

2.1.3. §343: How to Understand Our Cheerfulness

Here, Nietzsche further elaborates on the cultural consequences of the “death of God.” He writes that this event “has already begun to cast its first shadow over Europe.” For the few sensitive ones, “some kind of sun seems to have set; some old deep trust has turned into doubt.” But on the whole, this event “is still far too great, distant, and out of the way” for most people’s comprehension — people do not yet know “how much must collapse” as a result, including “our entire European morality.”

2.1.4. §344: In What Way We, Too, Are Still Pious

This is a crucial turning point in Nietzsche’s thought system. He points out: **Even science rests on a faith.** Science presupposes the rejection of all convictions — convictions have no right of citizenship in science, and must be downgraded to “hypotheses,” “tentative standpoints,” or “regulative fictions” to be accepted. However, Nietzsche asks: in order for the scientific spirit to begin, must there not be an “antecedent conviction” — one so authoritative and unconditional that it sacrifices all other convictions to itself? This antecedent conviction is: “Nothing is more necessary than truth.” The foundation of science is a **moral commitment** — an unconditional faith in the value of truth. And this faith itself has never been justified by science itself.

“Thus I deny morality as I deny alchemy, that is, I deny their premises: but I do not deny that there have been alchemists who believed in these premises and acted in accordance with them.”

2.2. Reading II: *Twilight of the Idols* (1888), “The Problem of Socrates” (pp. 162–6)

This section comprises §1 through §12. Here Nietzsche pushes his hostility toward Socrates — dating back to *The Birth of Tragedy* — to its extreme:

- §1–2: “Socrates is a symptom of decline” — “I recognized Socrates and Plato as symptoms of decay, as agents of Greek disintegration, as pseudo-Greek, as anti-Greek.”
- §3–4: Socrates’ ugliness is interpreted by Nietzsche as a physiological signal — he belongs to the “criminal type,” a bodily expression of the “décadent.”
- §5: Socrates elevated reason to supreme judge, creating the equation “reason = virtue = happiness.” Nietzsche sees this as anti-life — reason is not the source of life but its enemy.
- §8–10: Socrates “wanted to die” — he was not forced to drink poison by the Athenian trial, but “he himself forced the hemlock upon himself.” He was weary of life and used rational argumentation to find justification for his own death.

- **Core thesis: “Judgments, value judgments on life, for or against, can ultimately never be true.”** The living are an interested party and cannot be judges; the dead cannot judge either. Any judgment asserting that life has or does not have value is merely a physiological symptom, not truth.

Nietzsche's critique of Socrates is essentially a critique of **rationalism**: when reason is elevated to the point of suppressing instinct and life, it ceases to be an instrument of liberation and becomes a weapon of enslavement.

3. Supplementary Readings

3.1. *Ecce Homo*: How to Become What You Are

My campaign against morality begins with this book. Not that it has the slightest scent of gunpowder:-if you have some subtlety in your nostrils, you will smell very different and much pleasanter odours in it. Neither big guns nor small ones: if this book has a negative effect, its means are anything but that; the effect follows from these means like an inference, not like cannon fire. If you take leave of this book with a sort of timid caution towards everything that has been honoured and even adorned under the aegis of morality, this is not at odds with the fact that there is not a single negative word in the entire book, not a single attack or piece of malice, - that instead it lies in the sun, round and happy like a sea creature sunning itself between rocks. Ultimately, this is what I was, this sea creature: almost every sentence in the book was thought, hatched, in that jumble of rocks near Genoa where I was alone and still had secrets with the sea. Even now, whenever I happen to come into contact with this book, almost every sentence turns into a hook, pulling something incomparable from out of the depths: its whole skin trembles with gentle shudders of memory.

Nietzsche declares from the outset that he is launching a “campaign against morality.” But note: he is not opposing all morality, but rather the traditional morality rooted in Christianity and Platonism — the morality that treats “selflessness,” “compassion,” and “self-denial” as the highest values.

Daybreak is the first shot in this campaign. However, Nietzsche immediately emphasizes: there is no smell of gunpowder in this book. That is, his method of critique is not fierce attack or incitement to hatred, but a subtler, cooler, almost inference-like approach to dismantling old moral beliefs.

3.2. *Daybreak*: Thoughts on the Prejudices of Morality §103

There are two kinds of deniers of morality. - ‘To deny morality’ - this can mean, first: to deny that the moral motives which men claim have inspired their actions really have done so - it is thus the assertion that morality consists of words and is among the coarser or more subtle deceptions (especially self-deceptions) which men practise, and is perhaps so especially in precisely the case of those most famed for virtue. Then it can mean: to deny that moral judgments are based on truths. Here it is admitted that they really are motives of action, but that in this way it is errors which, as the basis of all moral judgment, impel men to their moral actions. This is my point of view: though I should be the last to deny that in very many cases there is some ground for suspicion that the other point of view - that is to say, the point of view of La Rochefoucauld and others who think like him - may also be justified and in any event of great general application. - Thus I deny morality as I deny alchemy, that is, I deny their premises: but I do not deny that there have been alchemists who believed in these premises and acted in accordance with them. - I also deny immorality: not that countless people feel themselves to be immoral, but there is any true reason so to feel. It goes without saying that I do not deny - unless I am a fool - that many actions called immoral ought to be avoided and resisted, or that many called moral ought to be done and encouraged - but I think the one should be encouraged and the other avoided for other reasons than hitherto. We have to learn to think differently - in order at last, perhaps very late on, to attain even more: to feel differently.

Nietzsche points out that there are two kinds of denial of morality:

The first kind (the position of La Rochefoucauld and others): holds that the moral motives people claim are not their real driving forces — morality is merely a set of words and deceptions (including self-deceptions), especially in the case of those most famed for virtue.

The second kind (Nietzsche's own position): admits that moral motives can indeed drive action, but these motives are based on errors rather than truths. That is, the foundations of moral judgment are fallacies, not objective realities.

Nietzsche denies morality in the same way he denies alchemy — denying its premises, but not denying that there were alchemists who believed in those premises and acted according to them.

The ultimate goal is: first learn to think differently, then learn to feel differently.

3.3. *The Gay Science* (1882/7) §343

How to understand our cheerfulness. – The greatest recent event – that ‘God is dead’; that the belief in the Christian God has become unbelievable – is already starting to cast its first shadow over Europe. To those few at least whose eyes – or the suspicion in whose eyes is strong and subtle enough for this spectacle, some kind of sun seems to have set; some old deep trust turned into doubt: to them, our world must appear more autumnal, more mistrustful, stranger, ‘older’. But in the main one might say: for many people’s power of comprehension, the event is itself far too great, distant, and out of the way even for its tidings to be thought of as having arrived yet. Even less may one suppose many to know at all what this event really means – and, now that this faith has been undermined, how much must collapse because it was built on this faith, leaned on it, had grown into it – for example, our entire European morality.

3.4. *The Gay Science* (1882/7) §344

In what way we, too, are still pious. – In science, convictions have no right to citizenship, as one says with good reason: only when they decide to step down to the modesty of a hypothesis, a tentative experimental standpoint, a regulative fiction, may they be granted admission and even a certain value in the realm of knowledge – though always with the restriction that they remain under police supervision, under the police of mistrust. But doesn’t this mean, on closer consideration, that a conviction is granted admission to science only when it ceases to be a conviction? Wouldn’t the cultivation of the scientific spirit begin when one permitted oneself no more convictions? That is probably the case; only we need still ask: in order that this cultivation begin, must there not be some prior conviction – and indeed one so authoritative and unconditional that it sacrifices all other convictions to itself? We see that science, too, rests on a faith; there is simply no ‘presuppositionless’ science. The question whether truth is necessary must get an answer in advance, the answer ‘yes’, and moreover this answer must be so firm that it takes the form of the statement, the belief, the conviction: “Nothing is more necessary than truth; and in relation to it, everything else has only secondary value.”

Summary: **The rejection of convictions itself depends on a stronger conviction; science, too, rests on a faith. Scientists, in their pursuit of truth, are in fact just like devout believers, holding an unshakable conviction.**

3.5. *Twilight of the Idols* (1888), ‘The problem of Socrates’, §2

This piece of irreverence, that the great sages are types of decline, first dawned on me in just the sort of case where scholarly and unscholarly prejudice would be working most strongly to prevent it: I recognized Socrates and Plato as symptoms of decay, as agents of Greek disintegration, as pseudo-Greek, as anti-Greek (*Birth of Tragedy*, 1872). The consensus sapientium – I see this with increasing lucidity – proves least of all that the wisest men were right about what they agreed on: instead, it proves that they were in physiological agreement about something, and consequently adopted – had to adopt – the same negative attitude towards life. Judgments, value judgments on life, for or against, can ultimately never be true: they have value only as symptoms, they can be taken seriously only as symptoms, – in themselves, judgments like these are stupidities. You really have to stretch out your fingers and make a concerted attempt to grasp this amazing piece of subtlety, that the value of life cannot be estimated. Not by the living, who are an interested party, a bone of contention, even, and not judges; not by the dead for other reasons. – It is an objection to a philosopher if he sees a problem with the value of life, it is a question mark on his wisdom, an un-wisdom. – What? So not only were the great sages all decadents but – they weren’t even sages? – But let me return to the problem of Socrates.

All judgments about the value of life are foolish, because the value of life cannot be objectively assessed — the living are interested parties who cannot serve as judges; the dead cannot judge at all. Any judgment asserting that life “has” or “does not have” value is merely a physiological symptom, not truth. To be puzzled about the value of life is a philosopher’s failure.

4. Revaluation of Values

Nietzsche characterizes his philosophical task as “Umwertung aller Werte” (the revaluation of all values). This is not merely a negation of existing morality — it is a **thoroughgoing re-examination of the very sources and foundations of value**.

- Old values (centered on Christian-Platonism): locate the “good” in a transcendent realm (God, the world of Forms), denigrating life, the body, instinct, and this-worldly existence
- New values: take **life itself** as the ultimate standard of value — whatever promotes the flourishing, creation, and self-overcoming of life is “good”; whatever suppresses life is “bad”

Revaluation means: we no longer accept any “given” authority of value — whether God, tradition, or reason — but **become creators of value ourselves**. This follows directly from the “death of God”: when external authority collapses, the task of creating values falls back upon humanity itself.

5. My Understanding

In short, Nietzsche is deeply passionate. His thought can be summarized as: **“Become who you are” — not discovering a predetermined self, but continuously creating a higher self through struggle, affirmation, and creation.**

To quote a passage: Nietzsche’s “original self” is social — it is the self we are forced to be shaped into and to become within our social roles, determined by our social nature, a “product” of the combined forces of environment, education, institutions, and culture. But “becoming who you are” means continuously challenging the “forces” of existing elements, constantly breaking through and exploring the self that has already been formed, persistently pursuing the inner “true self,” and continuously “awakening,” “developing,” “revealing,” and even, following one’s heart, “recreating” that objectively existing kernel self that is “at ease in its own nature”!

This, I believe, is the essence of Nietzsche!

Gottlob Frege

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6. Background

Gottlob Frege (1848–1925): German mathematician, logician, and philosopher, widely recognized as the founder of analytic philosophy and modern logic. His work fundamentally changed the direction of logic, the philosophy of mathematics, and the philosophy of language.

Frege spent almost his entire academic career at the University of Jena, but his influence spread widely through the inheritance and critique of thinkers such as Russell, Wittgenstein, and Carnap. His core ambition was **Logicism** — to demonstrate that all arithmetical truths can be reduced to purely logical truths.

7. Frege's Major Works

1873	On a Geometrical Representation of the Imaginary Forms in the Plane Doctoral dissertation
1879	Concept Script: A Formal Language of Pure Thought Modelled Upon That of Arithmetic The birth of modern logic
1882	On the Scientific Justification of a Concept Script
1884	The Foundations of Arithmetic: A Logico-Mathematical Enquiry Into the Concept of Number — the first systematic exposition of the logicist program
1891	Function and Concept
1892	On Sense and Meaning On Concept and Object — core texts of Frege's philosophy of language
1893-1903	Basic Laws of Arithmetic (two volumes) — the culmination of the logicist program
1910-13	Lectures on Concept Script
1918-23	Thoughts, Negation, Compound Thoughts — final reflections on the nature of logic and thought

8. Concept Script: The Birth of Modern Logic

8.1. Motivation: Overcoming the Deficiencies of Language

Frege's goal was to construct a completely precise, unambiguous formal language for purely logical derivation. He believed that natural languages (such as German, English) are riddled with ambiguity and imprecision, incapable of supporting rigorous logical reasoning.

“So that nothing intuitive could intrude here unnoticed, everything had to depend on the chain of inference being free of gaps. In striving to fulfil this requirement in the strictest way, I found an obstacle in the inadequacy of language: however cumbersome the expressions that arose, the more complicated the relations became, the less the precision was attained that my purpose demanded. Out of this need came the idea of the present Begriffsschrift.”

Thus, all truths that need to be proved can be divided into two categories:

- 1) Truths provable by logic alone (purely logical truths)
- 2) Truths that must rely on empirical facts to be proved (empirical truths)

Frege sought to determine: to which category do arithmetical truths belong? He attempted to reduce “the concept of ordering in a sequence” to “the concept of logical consequence,” thereby transitioning to the concept of “number” — allowing no intuition to intrude undetected at any point.

8.2. The Core Innovations of Concept Script

Frege’s Begriffsschrift is the first formal system of **predicate logic** in history. Its core innovations include:

- **Function-argument analysis:** decomposing propositions into functions and arguments, rather than the traditional subject-predicate structure. For example, “Socrates is mortal” is analyzed as: function $f(x) = “x \text{ is mortal}”$, argument $x = \text{Socrates}$
- **Quantifiers:** the first introduction of explicit notation for universal and existential quantifiers
- **Judgment stroke** and content stroke: distinguishing judged content ($\vdash$) from assertoric judgment ($\vdash\text{—}$)
- **Truth-values:** the referent (Bedeutung) of a sentence (proposition) is its truth-value (the True or the False), not an object

This makes it possible to check chains of inference **mechanically**, truly achieving “gapless logical derivation.”

9. Sense and Meaning (Sinn und Bedeutung)

This is Frege’s most far-reaching contribution to the philosophy of language. Published in 1892 as “On Sense and Meaning.”

9.1. The Problem: The Cognitive Value of Identity Statements

Frege begins with a deceptively simple puzzle:

What is the difference between “ $a = a$ ” and “ $a = b$ ”?

- “The Morning Star = the Morning Star” is a trivial analytic truth, providing no new information
- But “The Morning Star = the Evening Star” is an astronomical discovery! It provides new information

If the meaning of a name consisted solely in the object it refers to, then “The Morning Star = the Evening Star” should be just as trivial as “The Morning Star = the Morning Star” — but clearly it is not. Therefore, there must be something more than reference.

9.2. Frege’s Solution

Frege distinguishes three levels:

Reference (Bedeutung): the object itself to which the name refers. For sentences, the reference is a **truth-value** (the True or the False).

Sense (Sinn): the “mode of presentation” (Art des Gegebenseins) of the object. The same object can be presented in different ways. “The Morning Star” and “The Evening Star” have different senses — one presents as the star visible at dawn, the other as the star visible at dusk — but they refer to the same object (Venus).

Representation (Vorstellung): private, subjective mental images — irrelevant to logic.

“The sense of a proper name is grasped by everybody who is sufficiently familiar with the language or totality of designations to which it belongs; but this serves to illuminate only a single aspect of the reference, supposing it to have one. Comprehensive knowledge of the reference would require us to be able to say immediately whether any given sense belongs to it. To such knowledge we never attain.”

Key results:

- Sense determines reference
- The same reference can correspond to multiple senses
- Sense is public (graspable by all speakers), representation is private

9.3. The Paradox of Analysis

If for any “ $A = B$ ”, “ A ” and “ B ” either have the same meaning or they do not:

- If the meaning is the same, then “ $A = B$ ” is trivial
- If the meaning is different, then “ $A = B$ ” is incorrect

Frege’s solution: in “ $A = B$ ”, “ A ” and “ B ” can have different **senses** but the same **reference**. Thus “ $A = B$ ” is neither trivially true (because it provides new information about senses) nor incorrect (because the reference is the same).

10. Function and Concept: Frege’s Logical Ontology

10.1. Objects, Functions, and Concepts

Frege proposed a radical distinction:

Object: anything that is not a function. An expression for an object contains no empty place. — “I call anything a proper name if it is a sign for an object.”

Function: an unsaturated expression containing an empty place. E.g., “ $f(x) = 2x + 1$ ”

Concept: a special kind of function — one whose value is always a truth-value (True or False). For example, the function “___ is a table” maps each argument to the True or the False.

“A concept is a function whose value is always a truth-value.” (Frege, 1891)

10.2. Proper Name vs Concept Word

- **Proper name:** an expression referring to a single object (e.g., “the Moon,” “Mars”)
- **Concept word:** an expression referring to a concept (e.g., “*is human*,” “is an animal”)

Frege deliberately avoids the traditional term “common name” because it leads to the mistaken assumption that common nouns relate to objects in essentially the same way as proper names do. This is precisely what he rejects — concept words do not relate directly to objects; they relate through the “intermediary layer” of concepts.

“The word “common name” leads to the mistaken assumption that a common name is related to objects in essentially the same way as is a proper name ... that is why I prefer “concept word” to “common name”.”

11. The Concept Horse Paradox

This is one of the most famous and illuminating puzzles in Frege’s philosophy.

Frege wants to state a patently true fact: “The concept *horse* is a concept.”

But the problem is: the subject of this sentence — “the concept horse” — occupies the subject position. According to the grammar of natural language, an expression in subject position appears to refer to an **object**. However, if “the concept horse” refers to an object, then saying “this object is a concept” is self-contradictory — because objects and concepts are utterly distinct logical types.

“If I say ‘the concept horse is a concept’, then the first ‘concept’ (in ‘the concept horse’) is in fact an object, and the second ‘concept’ (the predicate) is the actual concept. There is an unavoidable linguistic distortion here.”

This is a profound insight: the very structure of language forces us to “reify” concepts into objects in order to talk about them. This paradox reveals the fundamental tension between natural language and logical structure — which is precisely the deep reason Frege constructed his Begriffsschrift.

12. Frege in Dialogue with Russell and Wittgenstein

12.1. Influence on Russell

Russell was deeply inspired by Frege, but also discovered a fatal flaw in Frege’s system — Russell’s Paradox:

- Consider the set of all sets that are not members of themselves
- Is this set a member of itself?
- If it is, then it must not be; if it is not, then it must be
- This paradox directly destroyed Frege's Basic Law V

When Russell wrote to Frege in 1902 informing him of this paradox, Frege was preparing the second volume of *Basic Laws of Arithmetic* for publication. He responded in an appendix with the famous words: "A scientist can hardly encounter anything more undesirable than to have the foundation collapse just as the work is finished."

12.2. Influence on Wittgenstein

The early Wittgenstein explicitly acknowledged "Frege's great works" in the Preface to the *Tractatus*. Wittgenstein inherited Frege's fundamental insights into logical analysis but arrived at different conclusions — particularly regarding the relationship between "sense" and "reference," and the status of logical constants.

13. The Living Character of Language: Waismann's Supplement

Friedrich Waismann (1896–1959), member of the Vienna Circle and interpreter of the later Wittgenstein, provided an important supplementary perspective to Frege's program of a rigorous formal language:

"One must indeed be blind not to see that there is something unsettled about language; that it is a living and growing thing, adapting itself to new sorts of situations, groping for new means of expression, for ever changing."

Frege sought to construct a perfect, fixed, unambiguous logical language; Waismann (and the later Wittgenstein) remind us that natural language itself is perpetually changing, adapting, and growing. The tension between these two insights constitutes a central dynamic of analytic philosophy.

Language is not a rigid logical system, but a living thing, forever
in flux.

14. Assigned Reading Analysis: On Sense and Meaning (1892), pp. 151–9

This paper is the core text of Frege's philosophy of language. The assigned pages (pp. 151–9) cover the first half of the paper, the crucial passages where Frege introduces and argues for the **distinction between sense and reference**.

14.1. The Paper's Opening: The Cognitive Puzzle of Identity Statements

Frege begins with a seemingly simple question: why can statements of the form " $a = b$ " sometimes provide valuable knowledge, while " $a = a$ " cannot?

- "The Morning Star = the Morning Star" — is a priori, trivial, uninformative
- "The Morning Star = the Evening Star" — is an astronomical discovery! It expands our knowledge

If the meaning of a name = its referent, then "The Morning Star = the Evening Star" should be just as trivial as "The Morning Star = the Morning Star" — but it is not. Therefore, names must possess something **beyond their reference**.

14.2. The Tripartite Distinction: Reference, Sense, Representation

On pages 152–4, Frege introduces the tripartite distinction:

- **Reference** (Bedeutung): the object itself that the name refers to. For sentences (propositions), the reference is a **truth-value** — the True or the False. This is Frege's original contribution: the reference of a sentence is not a "state of affairs" or a "fact," but a truth-value.
- **Sense** (Sinn): the mode of presentation (Art des Gegebenseins) of the object. Sense is **public** — graspable by anyone sufficiently familiar with the language. Sense contains the "epistemological path" to the referent.
- **Representation** (Vorstellung/idea): private, subjective mental images — irrelevant to logic. Different people have different mental images of "the Moon," but these belong only to the individual.

Core principle: **Sense determines reference**. The same reference can correspond to multiple senses (“The Morning Star” and “The Evening Star” both refer to Venus, but with different senses); a single sense may not determine a unique reference (though Frege holds that this should not occur in a perfected logical language).

14.3. Pages 156–8: The Reference of Subordinate Clauses

Frege further analyzes: in indirect speech (such as “Copernicus believed that the Earth is round”), the reference of the subordinate clause shifts from its truth-value to its **customary sense**. This opened the floodgates for later discussions of “propositional attitude contexts” and intensional contexts.

14.4. The Paper’s Philosophical Significance

- This is the first systematic distinction in the history of analytic philosophy between two semantic levels of linguistic expressions
- It provided the foundational framework for later semantic theories (from Tarski to Davidson to Kripke)
- It resolved the cognitive paradox of identity statements while preserving the core intuition of referentialism for logical analysis
- Its concept of sense provided an objective, publicly graspable anchor point for “meaning” — avoiding the pitfalls of psychologism

15. Summary: Frege’s Philosophical Legacy

Frege’s contributions form the “founding triangle” of analytic philosophy:

- 1) **Logic**: Invented predicate logic (quantifiers, function-argument analysis), making modern logic possible
- 2) **Philosophy of Language**: Distinguished sense and reference (Sinn und Bedeutung), laying the foundational layers of semantic theory
- 3) **Philosophy of Mathematics**: Proposed the logicist program — mathematical truths can be reduced to logical truths (though later proved by Gödel to be impossible to fully realize)

His deepest insight is perhaps this: **Language cannot be regarded as a transparent medium**. The grammatical form of natural language often conceals underlying logical form — and the task of philosophy is to reveal the latter.

“The sense of a proper name is grasped by everybody who is sufficiently familiar with the language or totality of designations to which it belongs... Comprehensive knowledge of the reference would require us to be able to say immediately whether any given sense belongs to it. To such knowledge we never attain.” — Gottlob Frege, “On Sense and Meaning” (1892)

Bertrand Russell

Notes compiled from the instructor's lecture materials; for knowledge sharing only; not for any commercial use

16. Background

Bertrand Russell (1872–1970): one of the core founders of 20th-century analytic philosophy. He made foundational contributions to logic, the philosophy of language, epistemology, the philosophy of mathematics, and political philosophy. His philosophical enterprise can be understood as an unrelenting pursuit of **logical clarity** and **precision**.

His major works include *Principia Mathematica* (1910–13, co-authored with Whitehead), “On Denoting” (1905), *The Problems of Philosophy* (1912), and “Knowledge by Acquaintance and Knowledge by Description” (1911). He was awarded the Nobel Prize in Literature in 1950.

Russell believed that **philosophy should imitate mathematics** — clarifying concepts through logical analysis to eliminate metaphysical confusion. Philosophical problems often stem from: grammatical confusion, ambiguous descriptions, and the mismatch between surface grammar and logical form.

17. Acquaintance and Description: The Epistemological Foundation

17.1. Knowledge by Acquaintance

Russell distinguishes two fundamentally different types of knowledge. **Acquaintance** is direct, non-inferential knowledge. The objects of our acquaintance include:

- **Sense-data:** such as a patch of red appearing in my visual field right now
- **Our own mental states:** such as the pain I am feeling at this moment
- **Universals:** such as “redness,” “similarity,” etc.
- **(Possible) the Self:** Russell holds an uncertain attitude on this point

Key feature: acquaintance is **direct** — there is no inferential intermediary between the knower and the known object.

17.2. Knowledge by Description

When we know something through a definite description, we need not have direct acquaintance with that object.

“Every proposition which we can understand must be composed wholly of constituents with which we are acquainted. (Russell 1911)”

This is Russell's **fundamental epistemological constraint principle**: any proposition we can understand must be composed entirely of constituents with which we are acquainted.

When we say “The F is G,” we may never have had direct contact with that object. For example:

- “The present King of France is bald”
- “The tallest student in the class”

We understand these propositions even when the object does not exist. This is because we are acquainted with the universals that compose the description (such as the properties “king,” “bald,” etc.).

17.3. Key Distinction

Acquaintance	Description
Direct relation	Mediated by properties
Requires no inference	Requires propositional form

Certain	Fallible
Logical atoms	Logical constructions

18. On Denoting: Russell's Theory of Descriptions

This is Russell's most famous philosophical contribution (published in 1905), acclaimed as "the paradigm of analytic philosophy."

18.1. The Problem: The Paradox of Non-Being

Consider the following statement:

- "The present King of France is bald"

According to ordinary intuition, this sentence seems to be meaningful (we can understand it), but it seems to be **false** — because France has no king.

But if this sentence is false, then its negation should be true:

- "The present King of France is not bald"

This also seems false — for there is no king who can be described as having hair or lacking it.

This problem concerns how language can talk about non-existent things. Meinong's theory holds that "the present King of France" **must in some sense exist** ("subsist"), otherwise we could not meaningfully talk about it. Russell found this conclusion absurd.

18.2. Russell's Solution: Quantificational Analysis

Russell points out that surface subject-predicate structure conceals true logical form. "The F is G" is actually a **quantificational structure** composed of three sub-propositions:

- 1) **Existence**: there is at least one thing with the property F
- 2) **Uniqueness**: there is at most one thing with the property F
- 3) **Predication**: that thing which has the property F also has the property G

Thus, "The present King of France is bald" is analyzed as:

- (1) There exists a King of France
- (2) There is at most one King of France
- (3) That King of France is bald

Because sub-proposition (1) is false, the entire conjunctive proposition is **false** — without invoking any mysterious non-existent entities!

18.3. Philosophical Significance

The Theory of Descriptions is the core embodiment of Russell's "logical constructionism":

- Definite descriptions are **not** referring expressions — they are quantificational structures
- "Non-existent objects" and other mysterious entities are eliminated (a "robust sense of reality")
- Surface grammar seriously misleads our understanding of logical form
- **The task of philosophy is to reveal genuine logical form through logical analysis**

"The names that we commonly use, like 'Socrates,' are really abbreviations for descriptions ... The only words one does use as names in the logical sense are words like 'this' or 'that.' (Russell 1918)"

19. Particulars and Universals

19.1. Particulars

Particulars are concrete, individual things. In Russell's strict logical sense:

- Instances of sense-data (such as “this patch of red”)
- They are **logically independent entities**
- Genuine logical proper names refer only to particulars — such as the demonstratives “this,” “that”

19.2. Universals

Universals are repeatable entities:

- **Properties** (such as “redness”)
- **Relations** (such as “greater than,” “before”)
- **Similarity**

Russell defends **realism about universals**: without universals, logic and mathematics would be impossible. This stands in direct opposition to nominalism.

20. Meaning and Denotation

Both Russell and Frege are concerned with the distinction between meaning and reference, but they diverge fundamentally:

20.1. Frege’s Approach

- Distinguishes Sinn (sense) and Bedeutung (reference)
- Sense is a public, objective abstract entity
- Sense determines reference
- There exists a “semantic hierarchy”: sign → sense → reference

20.2. Russell’s Approach

- Propositions are complexes composed of **real constituents**
- There are no hidden “sense-entities”
- Definite descriptions are not singular terms but quantificational structures
- Emphasizes **logical analysis** rather than semantic hierarchy

Key contrast:

- Frege → semantic hierarchy
- Russell → logical constructionism

21. The Multiple Relation Theory of Judgement

Russell’s early theory of judgement attempts to explain: what happens when we judge “Socrates is mortal”?

Russell holds that judgement is not a dyadic relation between a judger and an abstract proposition (as Frege thought), but rather a **multiple relation between the judger and several real constituents**:

- The judger (J) has a **multiple relation** with Socrates, mortality, and logical form
- There is no “proposition” as an independent abstract entity

This avoids the problem of ontological commitment to “false propositions” — if judgement is an attitude toward a proposition, then false judgement seems to imply that “false propositions” must exist. The multiple relation theory eliminates this difficulty.

22. Logical Atomism

Russell’s logical atomism is the final form of his philosophical system:

- **The world** is composed of logical atoms (particulars, simple universals)
- **Facts** are simple relations between atoms
- **Propositions** are logical pictures of facts
- **Logical form** is the manner in which atoms combine

- **The endpoint of logical analysis** is that which is simple and cannot be further analyzed

“The fundamental epistemological principle in the analysis of propositions containing descriptions is this: Every proposition which we can understand must be composed wholly of constituents with which we are acquainted. (Russell 1911)”

This closely resonates with the early Wittgenstein’s views in the *Tractatus* — the latter being deeply influenced by Russell.

23. The Value of Precision

Russell holds that **precision** itself has philosophical value:

- Philosophical problems often arise from grammatical confusion — the surface form of language does not match underlying logical structure
- The primary task of philosophy is not to construct grand metaphysical systems, but to **clarify confusion** (clarification)
- Logical analysis is the core tool for achieving this goal
- Philosophy is not a distinct “source of truth” different from science, but a continuation of the scientific spirit — treating more abstract problems with the same precision and rigor

24. Assigned Reading Analysis: Knowledge by Acquaintance and Knowledge by Description (1911), pp. 108–18

This paper is the core text of Russell’s epistemology. The assigned pages (pp. 108–18) cover the first third of the paper, establishing the fundamental distinction between acquaintance and description and drawing out its epistemological significance.

24.1. The Paper’s Opening (pp. 108–9): Statement of the Problem

Russell gets straight to the point: we often know propositions about “the so-and-so” to be true without knowing who or what the “so-and-so” is. For example: “I know that the candidate who gets the most votes will be elected, although I do not know who the candidate who gets the most votes is.” The problem is: in such cases, **what exactly do we know?** When the object of knowledge is merely described, what is the object of knowledge?

Russell then gives the classic definition of acquaintance:

“I say that I am acquainted with an object when I have a direct cognitive relation to that object, i.e. when I am directly aware of the object itself.”

Acquaintance = a direct cognitive relation = the relation in which an object is presented to the subject. This is not a relation of judgement but a relation of **presentation**.

24.2. The Kinds of Objects of Acquaintance (pp. 109–11)

On pp. 109–14, Russell systematically enumerates the kinds of objects with which we can be acquainted:

- 1) **Sense-data**: such as the objects I am directly aware of when I see a color or hear a sound. Sense-data are typically complex — a patch of color in the visual field contains spatial relations.
- 2) **Objects of introspection**: in self-consciousness, we seem to be directly aware of changing complexes containing cognitive and conative relations. “When I see the sun, I am often not only aware of the sun, but also aware of the fact that I **am seeing** the sun.”
- 3) **Universals**: such as “whiteness,” “diversity,” and other abstract objects. Direct awareness of universals is called “conceiving,” and the universal conceived is called a “concept.”
- 4) **The Self** (possible): Russell is cautious on this point — he considers it probable that he is acquainted with the self, but is reluctant to fully develop the argument in this paper.

24.3. The Epistemological Constraint Principle (pp. 116–18): The Core Theorem

On the basis of these definitions, Russell states his most famous epistemological principle:

“Every proposition which we can understand must be composed wholly of constituents with which we are acquainted.”

This is known as **Russell’s epistemological constraint principle**. Its corollaries include:

- If we can understand a proposition containing “Napoleon,” then “Napoleon” must be a constituent with which we are acquainted — but Napoleon is dead, and we cannot possibly be acquainted with him. Therefore, “Napoleon” is not, in the logical sense, a genuine name, but rather an **abbreviated description**.
- Genuine logical proper names can only be demonstratives such as “this,” “that” — they directly point to objects with which we are acquainted (sense-data).
- The “names” of ordinary language are merely informal abbreviations for descriptions.

On p. 118, Russell extends this principle further to judgement: “Whenever a relation of supposing or judging occurs, the terms to which the supposing or judging mind is related must be terms with which the mind is acquainted.” This means we cannot judge about constituents with which we are not acquainted — even when we appear to judge about “Napoleon,” we are actually judging about certain universals with which we are acquainted (those that compose the description of “Napoleon”).

24.4. The Paper’s Philosophical Significance

- This paper fundamentally transformed analytic philosophy’s understanding of “knowledge” — knowledge is no longer a unified “true justified belief,” but is essentially split into two basic types
- It provides the **epistemological foundation** for Russell’s Theory of Descriptions — the reason we need the analysis of descriptions is precisely that most of our knowledge is not acquaintance but is obtained indirectly through description
- It delineates the boundaries of “logical atomism”: the endpoint of logical analysis is those simple elements with which we can only be acquainted and which cannot be further analyzed

25. The Contemporary Significance of Russell’s Thought

Russell’s method remains crucial today:

- The core methodology of analytic philosophy — transforming philosophical problems into logical-linguistic analysis — stems directly from Russell
- Influence on AI and computer science: the Theory of Types and the formalization of logical systems
- Contribution to epistemology: the acquaintance vs. description distinction remains the basic framework for discussions of direct vs. indirect knowledge
- A lesson for metaphysics: do not be deceived by the surface form of language — “existence” is not a predicate, “nothing” is not a name

“Three passions, simple but overwhelmingly strong, have governed my life: the longing for love, the search for knowledge, and unbearable pity for the suffering of mankind.” — Bertrand Russell, *The Autobiography of Bertrand Russell*

Ludwig Wittgenstein

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26. Background

Ludwig Wittgenstein (1889–1951) is one of the most distinctive and influential philosophers of the 20th century. His philosophical career divides into two starkly different phases, each producing a landmark work:

- **Early** (1910s–1920s): *Tractatus Logico-Philosophicus* (1921), deeply influenced by Frege and Russell
- **Later** (1930s–1951): *Philosophical Investigations* (1953), a fundamental critique of his own early thought

Wittgenstein was deeply influenced by Frege and Russell — he explicitly thanked them in the Preface to the *Tractatus* — but his later work departed radically from the mainstream path of analytic philosophy.

The Preface to *Philosophical Investigations* states that the book's topics include “the concepts of meaning, of understanding, of a proposition and sentence, of logic, the foundations of mathematics, states of consciousness, and other things” — but its structure is not a linear argument; rather, it consists of “sketches of landscapes made in the course of long and meandering journeys... approaching the same, or almost the same, points from different directions.”

27. Early Thought: *Tractatus Logico-Philosophicus* (1921)

27.1. The World Is the Totality of Facts

The *Tractatus* begins with a series of succinctly numbered propositions:

“The world is all that is the case. The world is the totality of facts, not of things.”

Traditional philosophy holds that the world is composed of individual “things” (such as tables, apples). Wittgenstein points out that the **basic unit** of the world is not things, but “facts” — the state of affairs in which things stand in certain relations.

- A state of affairs (Sachverhalt) is a combination of objects
- Objects are simple, indecomposable — they are the **substance of the world**
- All possible facts together constitute “logical space”
- The actual world is merely that portion of logical space that is realized

For example, “the apple is on the table” is a fact, while “the apple” and “the table” are merely elements that compose the fact.

27.2. The Picture Theory of Language

Core view: **A proposition is a picture of reality.**

- Names in a proposition correspond to objects in the world
- The arrangement of names in the proposition (logical form) corresponds to the way objects combine in a state of affairs
- Therefore, language (in ideal circumstances) can “picture” the world — propositions and facts share the same logical form

But this also means: **Language can only picture what can be pictured.**

27.3. Drawing the Limit: Saying and Showing

The core aim of the *Tractatus* is to **draw a limit to thought and language**:

“Thus the aim of the book is to draw a limit to thought, or rather — not to thought, but to the expression of thoughts: for in order to be able to draw a limit to thought, we should have to find both sides of the limit thinkable (i.e. we should

have to be able to think what cannot be thought). It will therefore only be in language that the limit can be drawn, and what lies on the other side of the limit will simply be nonsense.”

Wittgenstein’s conclusion:

- **What can be said** (facts expressible by propositions) must be said clearly
- **What cannot be said** (ethics, aesthetics, religion, the meaning of life, logical form itself) must be passed over in silence

27.3.1. The Ladder Metaphor

“My propositions serve as elucidations in the following way: anyone who understands me eventually recognizes them as nonsensical, when he has used them — as steps — to climb up beyond them. (He must, so to speak, throw away the ladder after he has climbed up it.)”

Wittgenstein’s own propositions — including all the propositions in the *Tractatus* — belong to the category of “what cannot be said.” They are nonsensical, but they are the rungs that help the reader “see” this point.

27.3.2. The Aphorism of Silence

“What we cannot speak about we must pass over in silence.”

The correct method of philosophy: say only the propositions of natural science (what can be said clearly), and when someone wants to say something metaphysical, point out to them that certain signs in their propositions have been given no meaning.

28. Later Thought: *Philosophical Investigations* (1953)

28.1. From Logical Analysis to Language-Games

The later Wittgenstein completely abandoned his early Picture Theory. He no longer sought the “essence” or “logical form” of language, but instead studied the **actual ways** language is used in human life.

“This book is written for such men as are in sympathy with its spirit. This spirit is different from the one which informs the vast stream of European and American civilization in which all of us stand. That spirit expresses itself in an onwards movement, in building ever larger and more complicated structures; the other in striving after clarity and perspicuity in no matter what structure. (Philosophical Remarks, Foreword)”

Not pursuing vast, complex theoretical edifices, but only pursuing **clarity, perspicuity, seeing through**.

28.1.1. A Story: Is the Question Itself Reasonable?

When a child asks “Why don’t people in Australia fall off the edge of the earth?” Wittgenstein would not explain gravity or physical facts, but instead ask back: why don’t you fall off the paper? (turning a piece of paper upside down).

Wittgenstein is concerned first and foremost not with the answer, but with **whether the question itself is reasonable**. Many philosophical problems confuse us not because they are too profound, but because their very formulation is **confused**.

28.2. The Critique of Referentialism (PI §§1–43)

Referentialism holds that: the meaning of a word is the object it refers to. This was a shared premise of Frege and Russell (although they developed it in different ways).

Wittgenstein attacks this at the very opening of *Philosophical Investigations*:

- **Augustine’s picture of language**: every word has a meaning, and this meaning is the object it stands for. This is “a too simple picture of language.”
- Wittgenstein replaces this with the concept of the “language-game” — a language-game is “the whole, consisting of language and the actions into which it is woven” (PI §7).
- Different kinds of words are like different tools in a toolbox — hammer, saw, ruler, glue — they operate in different ways, not all in one way (“standing for an object”).

28.3. Meaning as Use

Wittgenstein proposed a revolutionary thesis:

“For a large class of cases of the employment of the word “meaning” — though not for all — this word can be explained in this way: the meaning of a word is its use in the language.”

“Meaning is use” is the core insight of the later Wittgenstein:

- Don’t ask what a word “stands for” — instead, look at **how it is used** in linguistic life
- The meaning of a word does not lie in some mysterious entity it refers to (Form, universal, sense-datum), but in **what function it performs** in our actual communicative practices
- “Meaning” is not an abstract object, but a **social practice**

Note Wittgenstein’s caution: “for a large class of cases … though not for all” — he is not giving a “theory of meaning,” but **describing** a way of looking at meaning.

28.4. Language-Games (PI §7, §23)

“Language-game” is the core conceptual tool of Wittgenstein’s later philosophy:

- It is not a theory, but an **object of comparison** — by likening language to a game, it displays certain features of language
- Different language-games have different rules, different aims, different participants
- There is no “essence” shared by all language-games
- The expression “language-game” is meant to highlight: **speaking language is part of an activity, or of a form of life**

28.4.1. The Variety of Language-Games (PI §23)

- Giving orders and obeying them
- Describing the appearance of an object, or giving its measurements
- Constructing an object from a description (a drawing)
- Reporting an event
- Speculating about an event
- Forming and testing a hypothesis
- Making up a story and reading it
- Play-acting
- Guessing riddles
- Telling jokes
- Requesting, thanking, cursing, greeting, praying

28.5. Family Resemblance (PI §§65–68)

Traditional philosophy (including the early Wittgenstein himself) sought to find the “common essence” of all language use — something shared by all instances of “language.”

Wittgenstein rejects this impulse. In its place comes the concept of **family resemblance**:

- Just as in a large family, member A and member B share similar eyes, B and C share a similar nose, C and D share a similar temperament — but there is no single feature shared by **all** members
- The various uses of language are the same — they resemble each other in overlapping and criss-crossing ways, but have no common essence
- “These phenomena have no one thing in common which makes us use the same word for all — but they are related to one another in many different ways”
- “Don’t think, but look!”

28.6. The Rule-Following Problem (PI §185)

This is one of the most profound and difficult sections in Wittgenstein's later philosophy.

28.6.1. The Problem Scenario

"Then we get the pupil to continue one series (say "+ 2") beyond 1000 — and he writes 1000, 1004, 1008, 1012.

We say to him, "Look what you're doing!" — He doesn't understand. We say, "You should have added two: look how you began the series!" — He answers, "Yes, isn't it right? I thought that was how I had to do it." — Or suppose he pointed to the series and said, "But I did go on in the same way". — It would now be no use to say, "But can't you see ...?" — and go over the old explanations and examples for him again."

The pupil says "I went on in the same way" — he has written 1000, 1004, 1008, 1012 — he thinks he has understood the rule. But how do we prove he is wrong? All the explanations we gave him were **finite examples**. Any finite set of examples can be "continued" in infinitely many ways.

28.6.2. Three Interpretations (from course notes)

Strong Realism / Platonism:

- "After 2, 4, 6, ... 996, 998, 1000 should come 1002, 1004, 1006..." is an **objective fact**
- Objection: there are infinitely many possible ways to continue a given sequence
- Possible response: successful evolution (evolution selected the correct inductive pattern)
- Rebuttal: this is realism, not strong realism

Constructivism / Anti-Realism:

- There is no objective fact determining whether the correct continuation of 996, 998, 1000 is 1002, 1004, 1006...
- The standard of correct application of any rule does not exceed the range of our limited human capacity to apply that rule
- The "correctness" of a rule comes from humans — use, practice, convention

Deflationism:

- The question that strong realism and constructivism attempt to answer is **fundamentally misguided**
- The question "what constitutes a rule or standard of correctness" does not need to be **solved**; it needs to be **dissolved/deflated**
- Rules do have the features we ordinarily take them to have — there is no need to seek a "deeper foundation" for them
- Objection: "begging the question"
- Possible response: reject the question itself
- Rebuttal: but is this really Wittgenstein?

28.6.3. Deeper Significance

The rule-following problem reveals:

- **There are no human-independent, eternal rules** — rules are constructed by humans; correctness comes from use, practice, convention
- **Correctness is limited, human-made, relative**, not absolute
- Wittgenstein is not **denying** the existence of correctness, but questioning the deep assumptions of the question "what makes a rule application correct"
- Ultimately, rule-following **does not rest on interpretation**, but on **practice and form of life**

28.7. The Private Language Argument (PI §243–)

Wittgenstein argues: a language that is **logically comprehensible to only one person** is impossible.

- If the meaning of a word lies in its use, and use is essentially a **public** practice (governed by rules, subject to correction), then "private rules" are a contradiction
- A "sensation word" meaningful only to oneself — without a public criterion of use — is not a word at all

- Therefore, the Cartesian “private inner realm” as the source of meaning is an illusion

29. From Early to Later: Key Tensions

Early (<i>Tractatus</i>)	Later (<i>Philosophical Investigations</i>)
Language is a picture of facts	Language is a toolbox — functionally diverse
Propositions and facts share logical form	There is no such thing as “logical form”
What cannot be said must be passed over in silence	Silence itself is a language-game
The task of philosophy is logical clarification	The task of philosophy is descriptive — “leave everything as it is” (PI §124)
There is an essence of language	There are only family resemblances — “Don’t think, but look!”
Meaning through reference	Meaning through use

30. Assigned Reading Analysis: *Philosophical Investigations* (1953), §§1, 2, 7, 23, 43, 65–68, 143, 185

Philosophical Investigations is the masterpiece of the later Wittgenstein. The eight assigned passages constitute the skeleton of its core argument.

30.1. §1: Augustine’s Picture of Language and “Five Red Apples”

Wittgenstein opens with a passage from Augustine’s *Confessions* that captures a seemingly natural but deeply misleading picture of language: every word has a meaning, and that meaning is the object the word stands for. He then imagines a scenario: a person goes to a shop with a slip marked “five red apples,” and the shopkeeper opens the drawer marked “apples,” consults the color sample for “red,” and counts out “five.” This simple scenario reveals that different words operate in **radically different ways** — “apple” is a name, “red” is a standard, “five” is an operation — and cannot be uniformly reduced to “standing for an object.”

30.2. §2: The Builders’ Language-Game

Wittgenstein imagines the most “primitive” language: a builder and his assistant use four words — “slab,” “pillar,” “block,” “beam” — to communicate. When the assistant hears a call, he brings the corresponding stone. This “complete primitive language” amounts to a whole language-game. It illustrates: language can function meaningfully without definitions, without meta-linguistic descriptions — meaning resides in use.

30.3. §7: The Definition of Language-Game

“I shall also call the whole, consisting of language and the actions into which it is woven, the “language-game.””

A language-game is **the whole interwoven fabric of language and activity**. This is not a theoretical definition but an “object of comparison,” meant to let us see the actual workings of language, rather than seeking some hidden essence.

30.4. §23: The Diversity of Language-Games

“But how many kinds of sentence are there? Say assertion, question, and command? — There are countless kinds: countless different kinds of use of what we call “symbols”, “words”, “sentences”. And this multiplicity is not something fixed, given once for all; but new types of language, new language-games, as we may say, come into existence, and others become obsolete and get forgotten.”

Wittgenstein enumerates a series of language-games: giving orders, describing appearances, reporting events, speculating, making up stories, play-acting, guessing riddles, telling jokes, requesting, thanking, cursing, greeting, praying... This list is open-ended. The key insight: **language does not have only one function, as logicians would have us believe.**

30.5. §43: Meaning as Use

“For a large class of cases of the employment of the word “meaning” — though not for all — this word can be explained in this way: the meaning of a word is its use in the language.”

This is the most famous sentence of the later Wittgenstein. But note his caution: “a large class of cases — though not for all.” He is not providing a new “theory of meaning,” but offering a **descriptive tool** — don’t ask “what is the meaning of X?” but look at “how is X used in linguistic life?”

30.6. §§65–68: Family Resemblance

Wittgenstein advances the argument through an imaginary dialogue: an interlocutor insists that there must be a “common essence” shared by all instances of “language.” Wittgenstein’s answer is: “Don’t think, but look!” — look at the overlapping and criss-crossing similarities that actually exist among the various ways language is used, like the similarities among a large family in eyes, nose, gait, temperament — “family resemblances.” There is no **single** feature common to all members, yet a network of overlapping and criss-crossing similarities suffices to support our use of the single word “language” to cover all these phenomena.

30.7. §143: Understanding = Mastery of a Technique

Understanding is not a “mental state” or “inner process,” but **mastery of a technique**. Wittgenstein imagines someone trying “in their head” to imagine the correct continuation of a melody or a number series — the issue is not “what goes on in their head,” but whether they can **actually do it**. The knowledge of understanding is a **practical ability**, not an inner “mental object.”

30.8. §185: The Rule-Following Paradox

“Then we get the pupil to continue one series (say “+2”) beyond 1000 — and he writes 1000, 1004, 1008, 1012.”

This is one of the most profound passages in the entire book. The pupil has a different “natural” understanding of “going on in the same way” from ours. No finite explanation can rule out **all** possible misunderstandings — for any finite explanation can itself be “understood differently.” This exposes a philosophical predicament: the rule itself cannot determine its own application. Wittgenstein’s response (developed in subsequent passages) is: at some point, explanation must give way to **practice** — we simply **act** in a certain way, and we naturally tend to continue in the same way. Rule-following is ultimately rooted in the agreement of **form of life**, not in the infinite justification of reason.

31. Wittgenstein’s Influence on Subsequent Philosophy

Wittgenstein’s influence spans multiple fields:

- **Analytic philosophy**: Redefined philosophical method — from theory construction to conceptual clarification
- **Philosophy of language**: Meaning as use — influenced Austin, Searle and others’ speech act theory
- **Philosophy of mind**: The Private Language Argument — a powerful challenge to Cartesian dualism
- **Sociology of knowledge**: The practice-based account of rule-following — provided resources for Bloor’s “Strong Programme”
- **Ethics**: Though little is said, the insight of “silence” and the concept of “form of life” deeply influenced contemporary metaethics

“The aspects of things that are most important for us are hidden because of their simplicity and familiarity. (One is unable to notice something — because it

is always before one's eyes.) The real foundations of their inquiry do not strike people at all. Unless that fact has at some time struck them. — And this means: we fail to be struck by what, once seen, is most striking and most powerful.” — Ludwig Wittgenstein, *Philosophical Investigations*, §129

Martin Heidegger

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32. Epigraph

“Das Aushalten in der Fragwürdigkeit erscheint als der einzige menschliche Weg, um die Dinge in ihrer Unerschöpflichkeit, d.h. Unverfälschtheit zu bewahren. (To hold out in this constant questioning appears as the only human way to preserve things in their inexhaustibility, i.e., without distortion.) — Heidegger, *Die Frage nach dem Ding* (*What is a Thing*)”

Only by constantly questioning, without forcing definitive answers, can we allow things to preserve their original richness and truth, undistorted by us. This is the core spirit of Heidegger's philosophy — philosophy is not about finding answers, but about **sustaining the questioning**.

Another famous remark: “Aristotle was born, worked, and died. Let us, therefore, turn to his thought.” This expresses Heidegger's rejection of biographism — what matters is not the philosopher's life story, but the thought itself.

33. Assigned Reading Analysis: Being and Time (1927), Introduction & §7 (pp. 49–63)

The assigned reading includes the Introduction to *Being and Time* and §7 (The Phenomenological Method), the programmatic passages in which Heidegger articulates his core question and method.

33.1. Introduction: Questioning the Meaning of Being (pp. 21–40 / H. 2–H. 15)

Heidegger's point of departure: **the question of the “meaning of Being” has been forgotten by philosophy for millennia**. Ever since Plato and Aristotle, metaphysics has discussed “that which is” (das Seiende), but it has never asked the more fundamental question that makes it possible for beings to **show themselves** as beings: **What does “Being” itself mean?**

Heidegger accomplishes three things in the Introduction:

- 1) **Argues for the necessity of re-posing the question of Being:** Being is treated as “the most universal concept” (hence seemingly needing no inquiry), “an indefinable concept” (hence seemingly uninquirable), and “a self-evident concept” (hence seemingly unnecessary to inquire into). It is precisely these three prejudices (universality, indefinability, self-evidence) that together have caused the question of Being to be systematically forgotten.
- 2) **Reveals the priority of Dasein:** Being is always the Being of some being. But different beings exist in different ways (a stone “is” as present-at-hand; a human “is” as to-be). Among these, **Dasein** has a special status — because Dasein is capable of **questioning its own Being**. Therefore, in order to understand the meaning of “Being,” one must first carry out an ontological analysis of **Dasein**.
- 3) **Articulates the task of “destroying” the history of ontology:** not negating the tradition, but loosening the hardened shell of concepts, allowing the forgotten original experiences within the tradition to be heard anew.

33.2. §7: The Phenomenological Method (pp. 49–63 / H. 27–H. 39)

§7 is the **methodological manifesto** of the entire book. Here Heidegger clarifies what he means by “phenomenology.”

33.2.1. A. The Concept of Phenomenon (pp. 49–52 / H. 28–H. 31)

Heidegger traces “phenomenon” (Phänomen) back to the Greek φαίνόμενον (“that which shows itself”).

- **Phenomenon** = that which shows itself in itself, within beings
- **Semblance** (Schein) = something may show itself as what it is **not**

- The “**phenomenon**” in **phenomenology** = that which precisely does **not** show itself initially and for the most part, but which is **co-disclosed** in that which does show itself. This is the **meaning of Being** itself — it is covered up in everyday experience, yet it is precisely what makes the showing of beings possible.

33.2.2. B. The Concept of Logos (pp. 52–5 / H. 32–H. 34)

The original meaning of λόγος (logos / discourse) is not “reason” or “judgment,” but “**to make manifest what one is talking about**” (apophainesthai). The function of discourse is to let something be seen **from itself**, through **showing**.

33.2.3. C. The Preliminary Concept of Phenomenology (pp. 55–63 / H. 34–H. 39)

“Phenomenology” = λέγειν τὰ φαινόμενα = **letting that which shows itself be seen from itself, just as it shows itself from itself**.

Heidegger emphasizes: phenomenology is not a “standpoint” or a “school” — it is merely a **method**. “Abandon the label ‘phenomenology’ — the only question is whether the investigation genuinely lets things show themselves.”

The relation between phenomenology and ontology:

- Ontology studies the meaning of Being
- But the meaning of Being is covered up in everyday life — it is precisely that which “does not show itself”
- Therefore, the only path to ontology is phenomenology — **through letting beings show themselves, to reveal the Being that is concealed in this showing**

33.3. Reading Guidance

These two parts of the assigned reading lay the foundation and blueprint for the entire edifice of Heidegger’s philosophy. The Introduction clarifies the “why” (why the question of Being must be re-posed), the “point of entry” (through the analysis of Dasein), and “how to proceed” (through phenomenological method). §7 then unfolds the methodological content of “phenomenology” — not reducing it to a technical procedure, but restoring it as a fundamental philosophical attitude: **letting things show themselves from themselves**.

34. Dasein

34.1. What Is Dasein?

Dasein (being-there) is the core concept Heidegger introduces to replace traditional philosophical concepts such as “consciousness,” “mind,” and “subject.” Heidegger uses Dasein to designate **the human being**, but not understood as a present-at-hand entity (like Descartes’ “thinking thing”), but as a **mode of Being**.

“By using Dasein as a replacement for “consciousness” and “mind,” Heidegger intended to suggest that an individual is in the world in the mode of “uncovering” and is thus disclosing other entities as well as itself. In other words, Dasein is the “there” — or the locus — of Being and thus the metaphorical place where entities “show themselves” as what they are.”

Key features:

- Dasein is not an inner “mind” opposed to an “external world”
- Dasein is from the very beginning **Being-in-the-world** (In-der-Welt-sein)
- There is no “pure consciousness” detached from the world — this is Heidegger’s fundamental critique of the entire modern philosophical tradition from Descartes onward
- The character of Dasein is not “I think” (cogito), but “to be” (Zu-sein)

The fundamental distinction between Dasein and other beings is this: Dasein is capable of **questioning** its own Being. A stone is, but it does not care about its Being; Dasein not only is, but **relates to** its Being.

35. Three Modes of Being

Heidegger distinguishes three fundamentally different modes of Being:

35.1. Readiness-to-hand (Zuhandenheit / Availableness)

This is **the most primordial relation between humans and the world** — one of Heidegger's most important discoveries.

Example: the hammer. When we are using a hammer, we are not conscious of the hammer itself — we are conscious of the task we aim to accomplish (driving a nail, making something). The hammer is in a state of “readiness-to-hand”: it is absorbed into our activity, becoming an extension of our body, not scrutinized as a separate “object.”

- In this state, the meaning of a thing resides in its **function**, not its **substance**
- Things “withdraw” into the background of use
- We do not first encounter a “thing” and then assign it a “meaning” — in use, we are already directly within the thing's meaning

Example: texting on a phone. When you are absorbed in sending a message, the phone itself disappears — you are communicating directly with your interlocutor.

35.2. Presence-at-hand (Vorhandenheit / Occurrentness)

When a thing **withdraws from use** — when the tool breaks, is missing, or gets in the way — it then appears as an “object.”

Example: the phone freezes. Suddenly, you are no longer communicating through the phone — the phone itself appears before you as a bulky, black, non-functioning object. You begin to “examine” it, “scrutinize” it.

Heidegger's crucial judgment: **Traditional philosophy and science have mistakenly taken this derivative state (presence-at-hand) as the primordial state of things.** When Descartes defined matter as “extended thing” (res extensa), he was describing not the primordial way we encounter the world, but a theoretical mode **abstracted from** the primordial experience of readiness-to-hand.

35.3. Existence (Existenz)

This is **the mode of Being peculiar to Dasein**. Unlike the “present-at-hand” Being of things, Dasein has no fixed essence — its essence is “to be.”

- Dasein continuously becomes itself through **choice and action**
- Dasein's possibilities are not “logical possibilities” but **existential possibilities** — these possibilities constitute Dasein's Being
- “The ‘essence’ of Dasein lies in its existence”

36. Totalities of Relevance

36.1. The Meaning of Things Is Not Isolated

Another crucial Heideggerian insight: **the meaning of a thing does not come from itself, but from its mutual references with other things.**

Hammer → nail → board → house → dwelling (shelter, family...) Phone → message → friend → relationship → community...

- **Totality of involvements** (Bewandtnisganzheit): the meaning of each tool depends on a whole network composed of other tools, purposes, and practices
- Without this network, a hammer would not be called a “hammer” — it would be merely a piece of wood attached to a piece of metal
- This totality is **prior** to individual things: we must first understand the meaning of the whole “workshop” before we can understand the meaning of the “hammer”

36.2. Pre-linguistic Reference vs Linguistic Reference

- **Pre-linguistic reference:** the natural associations in use that require no thought. The hammer “points to” the nail — this is not an inference we draw after thinking, but something we directly “see” in use
- **Linguistic reference:** the symbolization of meaning by words (e.g., the word “hammer” pointing to the thing hammer)

Heidegger holds that **pre-linguistic reference is more fundamental**. Language “parasites” on pre-linguistic, practical world-understanding. There is a deep resonance here with the later Wittgenstein.

Core thesis: “**In the encounter-structure of the world, what plays the primary role is not things, but references; not substances, but functions.**”

37. The They, Fallenness, and Authenticity

37.1. The They (das Man)

The everyday mode of Being of Dasein is that of **the They**:

- Conformity: living according to the standards of others, doing “what one does”
- Drifting with the current: not choosing one’s own mode of Being, but being shaped by social norms and expectations
- Losing oneself in “average everydayness”
- This state is called **Fallenness** (Verfallen)

“We take pleasure and enjoy ourselves as *they* take pleasure; we read, see, and judge about literature and art as *they* see and judge; likewise we shrink back from the “great mass” as *they* shrink back; we find “shocking” what *they* find shocking. The “they”, which is nothing definite, and which all are, though not as the sum, prescribes the kind of Being of everydayness. (Heidegger, *Being and Time*, §27)”

“The They” is not any concrete person — it is an anonymous authority that governs all domains of everyday life.

37.2. Authenticity (Eigentlichkeit)

Authenticity is “**owned existence**” — Dasein confronts its own Being, taking on its own choices and fate.

The key to authenticity is **Being-towards-death** (Sein-zum-Tode):

- Becoming aware of one’s own finitude and mortality
- Death is “one’s ownmost, non-relational, not-to-be-outstripped possibility”
- Precisely because death is irreplaceable (no one can die for you), it makes every Dasein an **individual**

“Authentic Being-towards-death is the willingness to be what I am right now, acknowledging that this has always been me.”

In the state of “the They,” people evade confronting death with the impersonal phrase “one dies.” But it is precisely through confronting one’s own mortality that Dasein can liberate itself from the dominion of “the They” and genuinely own its own life.

38. Mood and Language: Dasein’s Ways of Being Disclosed

38.1. Mood (Stimmung)

Heidegger holds that mood is not a “purely subjective” mental state, nor a reaction caused by external things. Mood is **a fundamental way in which Dasein discloses itself** — in a mood, we find ourselves already “there,” in the world, “touched” by the world in a certain way.

- Mood is “pre-reflective”: it is not the product of thought or judgment, but more primordial than reflection
- Boredom reveals the world as a whole lacking meaning; anxiety reveals the uncanniness and groundlessness of the world

38.2. Language (Rede / Discourse)

Heidegger holds that **language itself has the mode of Being of Dasein**. Language is not just another “object” appearing in the world — language is the **articulation of intelligibility**. We **live in** language, just as we live in the world.

39. Heidegger’s Relation to Western Modern Philosophy

Relatum	Points of Convergence	Fundamental Divergence
Descartes	Inherits the importance of the “self” as the point of departure for philosophy	Thoroughly rejects the isolated subject of “I think, therefore I am”: Dasein is from the very start in-the-world
Wittgenstein	Both oppose theoretical “representationalism” — opposing the view of knowledge as “the mind’s correct representation of the world”	Heidegger appeals to phenomenological description (“let things show themselves”), Wittgenstein appeals to the description of language-games
Murdoch	Both acknowledge moral possibilities beyond the level of convention; both focus on the importance of “attention” itself	Murdoch emphasizes love and the Good as objective reality; Heidegger emphasizes Dasein’s authentic choice
Sartre	Both take Heidegger’s analysis of Dasein as their starting point, treating “existence precedes essence” as the fundamental condition of the human	Heidegger rejected Sartre’s “humanist” reading; he is not talking about “human freedom,” but questioning the “meaning of Being”

40. Summary

Heidegger’s thought can be condensed into several layers:

- 1) **Methodology**: Phenomenology — letting things show themselves, rather than imposing theoretical frameworks upon them
- 2) **Critique of traditional philosophy**: The entire Western philosophical tradition from Plato through Descartes to Husserl commits the same fundamental error — confronting the world as an “object,” forgetting the more primordial “Being-in-the-world”
- 3) **Positive philosophical contributions**: the distinction between readiness-to-hand and presence-at-hand, totalities of relevance, the analysis of das Man / authenticity, “Being-in-the-world” as Dasein’s fundamental structure
- 4) **Ethical-existential meaning**: confronting death, taking on choice, emerging from the fallenness of das Man — becoming one’s authentic self

“The aspects of things that are most important for us are hidden because of their simplicity and familiarity. (One is unable to notice something — because it is always before one’s eyes.) The real foundations of their inquiry do not strike people at all. Unless that fact has at some time struck them. — And this means: we fail to be struck by what, once seen, is most striking and most powerful.” — Ludwig Wittgenstein, *Philosophical Investigations*, §129 (Hannah Arendt found these words deeply resonant with her own philosophy)

Susanne K. Langer

Notes compiled from assigned reading materials; for knowledge sharing only; not for any commercial use

41. Background

Susanne K. Langer (1895–1985): American philosopher, a central figure in 20th-century semiotic aesthetics. Her work spans the philosophy of logic, the philosophy of mind, and the philosophy of art. She was deeply influenced by Ernst Cassirer's philosophy of symbolic forms, while simultaneously inheriting and critiquing the theory of signs in the analytic philosophical tradition.

Core question: **Is art a symbol? If so, what kind of symbol is it?**

Major works:

- 1942: *Philosophy in a New Key: A Study in the Symbolism of Reason, Rite, and Art*
- 1953: *Feeling and Form: A Theory of Art*
- 1967–82: *Mind: An Essay on Human Feeling* (three volumes)

Langer explicitly positions *Feeling and Form* as the sequel to *Philosophy in a New Key* — the former established the foundations of symbol theory, the latter systematically applies it to art.

42. Core Problem and Method

42.1. The Mission of Philosophy

Langer inherits Charles Peirce's understanding of the task of philosophy:

“The business of philosophy is to unravel and organize concepts, to give definite and satisfactory meanings to the terms we use in talking about any subject (in this case art); it is, as Charles Peirce said, “to make our ideas clear.””

Langer explicitly states what her book does **not** do:

- Does not provide criteria for judging “masterpieces”
- Does not establish canons of taste
- Does not predict the boundaries and possibilities of art
- Does not teach anyone how to make art

These “norms and rulings” lie, in Langer's view, beyond the domain of philosophy. The task of philosophy is to **clarify concepts**, not to issue regulations.

42.2. The Foundations of Symbol Theory

Langer's philosophy is built upon the theory of symbols developed in *Philosophy in a New Key*. The core distinction:

Discursive Symbolism:

- Language and science belong to this category
- Possesses fixed structures of vocabulary and grammar
- Relations between symbols are logical and linear
- Capable of expressing “thoughts” but not “feelings”

Presentational Symbolism:

- Works of art belong to this category
- Does not possess a fixed “vocabulary” or “grammar”
- The symbol as a whole directly presents its meaning
- Capable of expressing “feeling” — i.e., the forms of human feeling

This is a fundamental distinction: **Art does not speak in an “artistic language”; art is an entirely different mode of symbolization.**

43. Art as the Form of Feeling

43.1. The Core Thesis

Langer's central claim: **Works of art are “presentational symbols”; they express not the artist's private feelings, but the “forms of feeling” — i.e., the logical structure of the life of feeling.**

Key distinction:

- **Expression vs. Self-expression:** When a person screams in anger, they are engaging in self-expression — releasing a feeling. But when an actor performs anger on stage, they are **expressing** anger — presenting anger as a form that can be perceived and understood.
- Art involves the latter, not the former.

43.2. The Meaning of “Feeling”

Langer uses “feeling” in an extremely broad sense, encompassing:

- Sensation
- Emotion
- Bodily awareness
- The affective tone of thought
- All qualities of experience that can be subjectively felt

Art expresses **the forms of these feelings** — tension and release, rising and falling, unfolding and contracting, conflict and resolution — not the feeling content of some specific person at some specific moment.

43.3. The Distinctiveness of Artistic Symbols

Langer emphasizes that artistic symbols (such as a piece of music, a painting) are essentially different from linguistic symbols:

- Linguistic symbols possess **fixed units of meaning** (words) and **rules of combination** (grammar); their meaning can be conveyed through translation
- Artistic symbols are **untranslatable** — the “meaning” of a painting can only be expressed by that painting itself
- The meaning of an artistic symbol is **inseparable** from its material carrier — change the sounds and you change the music; change the colors and you change the painting

“A work of art is a single, indivisible symbol, although a highly articulated one; it is not, like a discourse, composite, analyzable into more elementary symbols.”

44. Dialogue with Western Modern Philosophy

44.1. Relation to Wittgenstein

Langer's symbol theory is in deep dialogue with the early Wittgenstein:

- **Point of convergence:** Both insist on a rigorous boundary between “what can be said” (expressible in language) and “what cannot be said” (which can only be shown). Wittgenstein in the *Tractatus* says “What we cannot speak about we must pass over in silence”; Langer says that the unsayable can be **presented** through art, although it cannot be asserted.
- **Point of divergence:** Wittgenstein holds that “the unsayable” (ethics, aesthetics, the meaning of life) can only be shown, and any attempt to say it leads to nonsense; Langer holds that **art is precisely a systematic way of “showing”** — it is not nonsense, but a different mode of meaning.

44.2. Relation to Heidegger

Langer and Heidegger share certain deep concerns:

- Both oppose reducing all meaning to “assertoric truth” or “propositional knowledge”

- Both hold that art is a way in which truth happens — although Langer (influenced by Cassirer) uses the concept of “symbol,” while Heidegger uses the concept of “Being”
- Both are concerned with how “world” is “opened up” through symbols/language/art

The difference lies in: Langer maintains the clarity and systematicity of analytic philosophy; her “symbol” theory is far more precise and operational than Heidegger’s “truth of Being.”

44.3. Critique of Logical Empiricism

Langer’s symbol theory constitutes a powerful challenge to the logical empiricist “criterion of meaning.” If only “discursive symbolism” (i.e., scientific language) is considered “meaningful,” then:

- The entire domain of art is either excluded from meaning (relegated to “emotional discharge”) or forcibly reduced to some secondary form of discourse
- The dimensions of human experience that can **only** be expressed through presentational symbols — the architectonic feelings of music, visual art, dance — are erased by philosophy

Langer restores art to its fundamental place in the human system of meaning: **Art is not the poor cousin of science, but an autonomous, fundamental, and irreplaceable mode of symbolization.**

45. Assigned Reading Analysis: Feeling and Form (1953), Chapter 1: “The Concept of the Symbol”

Chapter 1 is the cornerstone of Langer’s theory of artistic symbols. In this chapter, she accomplishes two important tasks: delineating the scope and philosophical mission of the book, and establishing the basic conceptual framework of “symbol.”

45.1. The Task of Philosophy and the Limitations of This Book (Introduction, pp. vii–xii)

Langer makes clear in the Introduction that this book is the sequel to *Philosophy in a New Key* (1942) — it presupposes the reader’s familiarity with the symbol theory developed in that earlier work.

She clearly lists what this book **does not** do:

- Does not provide criteria for judging “masterpieces”
- Does not establish canons of taste
- Does not predict what is possible or impossible in art
- Does not teach anyone how to make art
- Does not reconcile art theory with any “world hypothesis” (such as those proposed by Stephen Pepper)

She writes: “All these norms and rulings, I think, fall outside the philosopher’s province. The business of philosophy is to unravel and organize concepts, to give definite and satisfactory meanings to the terms we use in talking about any subject.”

45.2. The Core of Chapter 1: Definition and Classification of the Symbol

Chapter 1 establishes the cornerstone of Langer’s entire aesthetic theory — the concept of the **symbol**. Langer distinguishes:

Sign: Indicates the existence or imminent occurrence of something — animals also respond to signs (smoke is a sign of fire). Signs have a direct, stimulus-response function.

Symbol: Not only indicates an object, but is used to **think about** the object. A symbol is the **vehicle for the conception of objects** — humans not only react to “smoke,” but can also **think about** fire, “talk about” it when fire is absent. This is precisely the dividing line between the human mind and the animal mind.

45.3. Discursive Symbolism

Language is the paradigmatic discursive symbolic system. Its features include:

- **Vocabulary:** fixed, conventional units of meaning (words), each with relatively determinate meaning
- **Grammar:** rules of combination that determine the legitimate ways complex expressions can be formed
- **Translatability:** statements in a discursive symbolic system can be translated into equivalent expressions in the same language or into another language

45.4. Presentational Symbolism

Langer's breakthrough contribution lies here: she argues that **there exists another, fundamentally different mode of symbolization** — presentational symbolism. This kind of symbol:

- **Has no fixed vocabulary or grammar** — a painting is not composed of “minimal units of meaning” assembled according to rules
- **Bears meaning holistically** — its parts cannot be detached and independently defined like words in a sentence; any local change alters the meaning of the entire work
- **Is untranslatable** — the meaning of a painting cannot be fully conveyed in words (or in any other medium)
- **Expresses the forms of the life of feeling** — not the content of feeling (“I was sad yesterday because...”), but the structure and dynamics of feeling (tension and release, rising and falling, unfolding and contracting, conflict and resolution)

45.5. “Significant Form”

Langer inherits Clive Bell's concept of “significant form” but provides it with a precise semiotic foundation. A work of art is a kind of “significant form” — a symbolic form that directly presents the logical structure of the life of feeling. It is not a statement **about** feeling (like a psychology paper), but a **symbolic whole** that **presents** the form of feeling.

“A work of art is a single, indivisible symbol, although a highly articulated one; it is not, like a discourse, composite, analyzable into more elementary symbols.”

45.6. Reading Guidance

Chapter 1 establishes the methodological principles and conceptual tools for the entire book. Langer's ambition is grand: she attempts to demonstrate that art is not a mysteriously ineffable “irrational” domain, but, like science and language, is a **fundamental, autonomous, irreplaceable** mode of the human symbolic capacity. In dimensions that science cannot reach — the forms of the human life of feeling — art is the sole mode of symbolization.

46. Summary: The Philosophical Significance of Langer's Thought

- 1) **Expanded the concept of “meaning”:** Meaning is not limited to propositions (true/false judgments of “X means Y”), but also encompasses the forms of feeling expressed through presentational symbols that are untranslatable into discursive language
- 2) **Art is cognitive:** Art is not “aesthetic diversion” or “emotional release”; it is a fundamental way for humans to know themselves and the world — it lets us **see** the forms of the life of feeling
- 3) **Systematization of symbol theory:** Langer inherited Cassirer's thesis of man as “animal symbolicum,” concretizing it into a theory of different modes of symbolization
- 4) **Bridging analytic and continental philosophy:** She embraces the analytic philosophical pursuit of clarity and precision, while also responding to continental philosophy's (especially Heidegger's) concerns about art, feeling, and non-propositional meaning

“The aim of philosophy is to make our ideas clear. But what does it mean to make an idea of art clear? Not to reduce art to a formula, but to understand what kind of phenomenon we are dealing with when we make or behold a work of art.” — Susanne K. Langer, *Feeling and Form*

Kurt Gödel

Notes compiled from the instructor's lecture materials; for knowledge sharing only; not for any commercial use

47. Background

Kurt Gödel (1906–1978): Austrian-American logician and mathematician, most famous for his **Incompleteness Theorems** (1931). He was not only a mathematician but also a thinker with deep philosophical interests, particularly concerned with formal systems, logic, mathematical truth, and ontological questions.

Gödel grew up in the logical and philosophical tradition of the Vienna Circle, deeply influenced by Frege, Russell, and Cantor. Yet he ultimately arrived at a philosophical position diametrically opposed to the Vienna Circle mainstream (logical empiricism).

What was the historical context for his incompleteness theorems? Frege's logicist program, Russell and Whitehead's *Principia Mathematica*, Hilbert's formalist program — all of these pursued the same great ambition: **to provide mathematics with an absolutely rigorous, unshakable logical foundation**. Frege believed all mathematical truths could be reduced to purely logical truths; Russell sought to repair the defects in Frege's system (Russell's Paradox) through type theory; Hilbert proposed using finite, constructive reasoning to prove the completeness and consistency of formal mathematical systems. Gödel proved: this goal is unattainable.

48. Assigned Reading Analysis: On Formally Undecidable Propositions (1931), pp. 173–9

The assigned pages (173–9) constitute the opening portion of Gödel's 1931 paper, one of the most important mathematical-philosophical papers of the 20th century.

48.1. The Structure and Starting Point of the Paper

Gödel opens by clearly stating his core objective: to prove that in *Principia Mathematica* (PM) and related systems, **there exist formally undecidable propositions** — i.e., propositions such that neither they nor their negations can be proved within the system.

The assigned pages cover:

- 1) A preliminary description of formal systems (particularly PM)
- 2) The basic idea of the Gödel numbering scheme
- 3) The strategy for the arithmetization of meta-mathematical concepts

48.2. Core Argumentative Strategy

Gödel's methodological innovation is: **mapping meta-mathematical statements about a formal system onto arithmetical statements within the formal system itself**.

- Each symbol, each string of formulas, each proof sequence is assigned a unique natural number (Gödel number)
- Meta-mathematical statements about provability ("proposition P is provable in system S") are translated into arithmetical statements about natural numbers
- Gödel then constructs a proposition G that is arithmetically equivalent to "I am not provable in system S"
- If S is consistent, then G is not provable in S (otherwise S would prove a false proposition); but G precisely states its own unprovability — hence G is true

48.3. The Impact on the Foundations of Mathematics

This paper directly destroyed Hilbert's formalist program:

- Hilbert hoped to use a "finitary," "constructive" meta-mathematics to prove the consistency and completeness of mathematical systems

- Gödel proved: any consistent formal system sufficiently strong to contain arithmetic cannot prove its own consistency (the Second Incompleteness Theorem)
- There is an unbridgeable gap between truth and provability — **some true propositions are unprovable in any given formal system**

48.4. Philosophical Corollaries of the Paper

Gödel himself drew clear Platonist conclusions from this mathematical result:

- Mathematical truth cannot be identified with “provability in a formal system”
- Mathematical objects (such as natural numbers, sets) have an objective existence independent of our formal systems
- The human mind is capable of “seeing” certain mathematical truths that formal systems cannot prove — this suggests a fundamental difference of some kind between mind and machine

It is worth noting that Gödel’s formulations in this paper are extremely cautious and technical — he carefully distinguishes the philosophical corollaries of his results from the mathematical content. It was only in his later “Gibbs Lecture” (1951) that he explicitly articulated the philosophical position implicit in the paper.

49. Gödel’s Incompleteness Theorems

49.1. Three Informal Formulations of the Theorems

The First Incompleteness Theorem can be stated in three ways:

- 1) “**No axiomatization can determine the whole truth and nothing but the truth concerning arithmetic.**”
- 2) “**Any axiomatization of arithmetic must either be incomplete — fail to capture some arithmetical truths — or be unsound — capture some arithmetical falsehoods.**”
- 3) “**No theory can be sufficiently strong, consistent, complete, and axiomatizable.**”

49.2. Gödel Numbering

The technical core of Gödel’s proof is Gödel numbering: assigning each expression in a formal language a unique natural number, such that:

(i) No two distinct expressions are assigned the same natural number (ii) There exists an algorithm that can determine what natural number is assigned to any given expression, and also determine what expression (if any) is assigned to any given natural number

Through Gödel numbering, Gödel enabled a formal system to “talk about itself” — analogous to a program being able to process its own source code. This is the mathematical foundation of self-reference.

50. Mathematical Platonism

Gödel’s core philosophical position is **mathematical Platonism**:

- **Mathematical objects exist objectively**: Numbers, sets, and propositions are not human inventions, but objective realities independent of human thought
- **Mathematical truths are “discovered” rather than “created”**: The mathematician’s work is more like an explorer discovering new continents, not a poet creating new imagery
- **Even mathematical propositions unprovable within one formal system can still be true**: Provability does not equal truth

This stands in stark contrast to formalism (such as Hilbert’s) and logical empiricism (the Vienna Circle mainstream). For Gödel, the incompleteness theorems precisely proved his Platonism: formal systems are merely limited “tools” that cannot fully capture the totality of mathematical truth.

50.1. The Absoluteness of Logic and Knowledge

Gödel was deeply influenced by Frege, holding that:

- Logical laws are not conventions — they exist independently of human thought
- Logical truths, like mathematical truths, are “discovered” rather than “invented”
- “The human mind is incapable of formulating all its mathematical intuitions” — whenever we succeed in formalizing some of them, this very fact yields new intuitive knowledge (this may be called the “incompleteness” of mathematics)

51. Common Misunderstandings of the Incompleteness Theorems

Misunderstanding 1: “Gödel’s theorem means that some arithmetical truths are unprovable.” — In fact: Gödel’s theorem concerns provability **within a given axiomatic system**, not unprovability in an absolute sense. A proposition unprovable in one system may be provable in another, stronger system.

Misunderstanding 2: “Gödel’s theorem applies to all domains.” — In fact: The incompleteness theorems specifically concern arithmetic (or any system sufficiently strong to contain arithmetic); they do not apply to all domains.

Misunderstanding 3: “Gödel’s theorem proves the existence of God.” — In fact: This is a serious misreading of Gödel’s thought. Gödel himself did indeed have an independent “ontological proof,” but this is an entirely different topic from the incompleteness theorems.

Misunderstanding 4: “Gödel proved that there exist arithmetical propositions asserting their own unprovability.” — In fact: This is an oversimplification.

52. Gödel and AI: The Philosophical Lineage of the Mathematical Objection

52.1. From Gödel to Turing to Lucas

The course lectures traced an important philosophical lineage:

Cantor’s work (the diagonal argument) established the concept that “there are different orders of infinity” — this is the mathematical foundation for all subsequent “self-reference” arguments.

Gödel’s incompleteness theorems → any sufficiently strong formal system contains undecidable propositions

Turing transformed this into the Mathematical Objection concerning machines: “There are certain things that a machine cannot do. If it is rigged up to give answers to questions, there will be some questions to which it will either give a wrong answer, or fail to give one at all…”

Lucas (1961) argued in “Minds, Machines and Gödel”:

“However complicated a machine we construct, it will correspond to a formal system, which in turn will be liable to the Gödel procedure of finding a formula unprovable-in-that-system. This formula the machine will be unable to produce as being true, although a mind can see that it is true. … Thanks to Gödel’s theorem, the mind always has the last word.”

Turing’s response (1950, anticipating Lucas’s argument):

- Humans themselves often give wrong answers — we have no grounds for complacency about machines’ fallibility
- Human superiority can only be demonstrated against one particular machine — it is impossible to triumph simultaneously over all machines
- “There might be men cleverer than any given machine, but then again there might be other machines cleverer again, and so on”

52.2. Incompleteness from a Modern AI Perspective

In the era of contemporary large-model AI, this discussion becomes subtler:

- Large models do not rely on fixed axioms or logical deduction — they “predict” outputs through statistical learning and pattern generation
- The question is no longer “can a formal system simulate a human,” but “can a statistical machine produce genuine understanding”

- Gödel's deepest philosophical impact is perhaps revealing the **boundaries of formalized rationality**: any closed system cannot encompass the totality of truth, and understanding may forever exceed the scope of rules and algorithms

53. The Gödel Universe: Time and Ontology

In his later years, Gödel also took an interest in physics and philosophy, especially the philosophy of time. He proposed the famous **Gödel Universe solution** — a solution to Einstein's field equations that permits closed timelike curves.

This touches on profound philosophical questions:

- If time travel is logically possible, what is the nature of time?
- Does existence have an objective structure independent of our temporal experience of it?
- This reflects Gödel's consistent Platonism: the physical world (like the mathematical world) has an objective structure independent of human experience

54. Gödel in Comparison with Western Modern Philosophical Traditions

Scholar/ School	Period	Core View	Relation to Gödel
Frege	1848–1925	Mathematical logicism; mathematics reducible to logic; logic is absolutely objective	Gödel was deeply influenced by Frege, inheriting logicist ideas, but emphasized mathematical Platonism and incompleteness
Russell	1872–1970	Logicism, discoverer of set-theoretic paradox; mathematics' foundation depends on logic and set theory	Gödel used the background of Russell's type theory in his incompleteness theorems, but transcended the limitations of formalized systems
Hilbert	1862–1943	Mathematical formalism; pursued system completeness and consistency	Gödel directly challenged Hilbert's formalist ideal: the incompleteness theorems show that mathematical systems cannot be completely complete
Vienna Circle	Early 20th c.	Logical empiricism; mathematics and logic determined by experience or linguistic analysis	Gödel opposed logical empiricism: mathematical truth is independent of experience; emphasized objective existence
Cantor	1845–1918	Founder of infinite set theory; objective infinity exists in mathematics	Gödel was influenced by Cantor, recognized the objective existence of mathematical objects, especially sets and mathematical objects
Wittgenstein	1889–1951	Philosopher of language; early emphasis on the limits of logical language, later emphasis on language use	Gödel criticized Wittgenstein's "linguistic limitation" view of mathematical truth, holding that truth is independent of language

Turing	1912–1954	Founder of computation theory; proposed the universal Turing machine and the Turing test	Gödel’s theorem is the basis of Turing’s “mathematical objection” concerning machine intelligence; Turing himself was critical of Lucas-style arguments
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55. Summary

Gödel’s philosophical legacy can be summarized as:

- 1) **Revealing the boundaries of formalized rationality:** Any closed formal system cannot encompass the totality of mathematical truth — “truth does not equal provability”
- 2) **Mathematical Platonism:** The mathematical world is an objective reality that humans discover, not invent
- 3) **Profound influence on AI and the philosophy of mind:** The incompleteness theorems sparked an ongoing debate about “whether the human mind can be mechanized”
- 4) **A fatal blow to logicism and formalism:** The grand ambitions of Frege, Russell, and Hilbert were proved impossible in principle

“The human mind is incapable of formulating ... all its mathematical intuitions, i.e., if it has succeeded in formulating some of them, this very fact yields new intuitive knowledge ... This fact may be called the ‘incompleteness’ of mathematics.” — Kurt Gödel, 1951 Gibbs Lecture

Hannah Arendt

Notes compiled from the instructor's lecture materials; for knowledge sharing only; not for any commercial use

56. Brief Introduction

56.1. Overview

Arendt has a unique philosophical perspective, as she even refuses to call herself a “political philosopher” and prefers to be called a “political theorist”. She believes that traditional political philosophy (since Plato) has always tried to regulate and even suppress the accidental nature of political life with metaphysical absolute truths. Her core concern is: after experiencing the disasters of 20th-century totalitarianism (Nazism and Stalinism), how should humanity rethink politics, defend freedom, and rebuild public life?

56.2. Who Was Arendt

Born in 1906 to a Jewish family in Hanover, Germany, she moved with her family to Königsberg at age three. She lost her father at seven and grew up in a secular and politically progressive environment. In 1914, fleeing war, she went to Berlin and completed her secondary education.

In 1924 she entered the University of Marburg, studying under Heidegger and Jaspers; she had a four-year romantic relationship with Heidegger. In 1929 she received her doctorate in philosophy under Jaspers' supervision, but could not teach at German universities due to her Jewish identity. That same year she married Günther Stern.

In 1933 she was arrested for researching anti-Semitism; after her release she fled Germany, settling in Paris via Czechoslovakia and Switzerland. She worked for Jewish organizations and befriended Walter Benjamin. In 1937 she was stripped of German citizenship; in 1940 she married Heinrich Blücher. In 1941, with help from an American diplomat, she fled through France and Spain to the United States, becoming a U.S. citizen in 1950.

In 1951 she published *The Origins of Totalitarianism*, which established her reputation. She subsequently authored *The Human Condition*, *Eichmann in Jerusalem*, and *On Revolution*. In 1959 she became Princeton University's first female full professor. She died of a heart attack in 1975 at age 69.

57. The Origins of Totalitarianism and the “Banality of Evil”

57.1. The Nature of Totalitarianism

In *The Origins of Totalitarianism*, Arendt argues that totalitarianism differs from traditional despotism or dictatorship. Traditional dictatorship merely restricts people's political freedom, whereas totalitarianism seeks to thoroughly destroy human “plurality” and “spontaneity,” controlling people's public and private lives in every dimension.

- **Ideology and Terror:** Totalitarianism relies on an ideology of “absolute historical laws” (such as Nazi racial superiority doctrine), and implements this logic through ubiquitous terroristic rule.
- **Atomized Society:** The precondition for the success of totalitarianism is the breakdown of bonds between people in modern society. People become lonely, indifferent, mutually isolated — becoming “atomized individuals” — and are thus easily incited and manipulated by totalitarian propaganda.

57.2. The Banality of Evil

In 1961, Arendt traveled to Jerusalem to report on the trial of Nazi war criminal Adolf Eichmann (the functionary responsible for sending millions of Jews to concentration camps). In *Eichmann in Jerusalem*, she proposed the famous concept of the “banality of evil.”

- **The Evil of Ceasing to Think:** Contrary to all expectations, Eichmann was not a fanatical demon, but a “terrifyingly normal person.” The root of his monstrous crimes was not bloodlust, but a “loss of the capacity to think” (Thoughtlessness). He blindly obeyed orders, treated bureaucratic logic as the highest principle, and lost the ability to think from another person's standpoint.

- **Implication:** Evil need not arise from extremely malevolent motives. When ordinary people in a society abandon independent thought and moral judgment, merely functioning mechanically as cogs in a system, the greatest catastrophes can unfold.

58. The Human Condition and the Threefold Structure of “Active Life”

To answer the question “How can humanity emerge from the shadow of totalitarianism?”, Arendt, in her magnum opus *The Human Condition* (also translated as *Vita Activa*), returned to ancient Greece and re-examined the three fundamental modes of human activity on earth. This constitutes the central pillar of her philosophical system:

58.1. Labor

It is cyclical and never-ending. What labor produces (food, energy) is quickly consumed. Labor subjects humans to the necessity of nature; at this level, humans are essentially no different from animals (Arendt calls this “Animal Laborans”).

58.2. Work

It has a clear beginning and end, and it leaves behind durable material results. Work transforms natural materials into civilizational tools, providing a relatively stable, intergenerationally transmissible “world” for human life. At this level, humans are “Homo Faber” (the fabricating human).

58.3. Action

- Plurality: Action must take place in a public space constituted by a plurality of people. Human beings live together on the earth, and each person is unique.
- Initium / Beginning: Every “action” is the injection of a wholly new, unpredictable force into the world. It means breaking with routine and creating new possibilities.

59. The Philosophy of Power: Its Essential Difference from Violence

Most political philosophers (such as Max Weber, Hobbes) hold that power is a form of coercive force — the ability to compel obedience backed by violence. But Arendt proposes a subversive view: Power and violence are opposites.

- Power belongs to the group: Power never belongs to any single individual. Only when people gather together and act in concert on the basis of mutual trust and consent does power come into being. Power is the manifestation of cohesion. Once the group disperses, power vanishes.
- Violence depends on tools: Violence is instrumental (such as firearms, armies). When rulers lose the consent of the people (lose “power”) and must rely on sheer violence to maintain rule, this precisely demonstrates that they have lost genuine power.

Conclusion: Totalitarian rule appears to possess the mightiest state apparatus, but in Arendt’s view, it is the most devoid of “power,” because it isolates everyone, utterly destroying the basis for people’s common action, relying solely on terror and violence to rule.

60. Assigned Reading Analysis: Eichmann in Jerusalem (1963–65), Postscript

The assigned reading is the Postscript Arendt added to the reissue of *Eichmann in Jerusalem* — the most important text for understanding the book’s controversies and Arendt’s core position.

60.1. Core Content of the Postscript

60.1.1. 1. Response to the Controversy and Clarification of Misunderstandings

Arendt opens by noting that this book had become the target of “an organized campaign” even before its publication. Many who attacked the book admitted they “had not read it and never would.” Arendt’s response is calm and sharp:

The controversy focused on distorted questions (which Arendt says she “never thought about”):

- Whether the Jews themselves should bear responsibility for the catastrophe? (A question Arendt calls “foolish and cruel”)

- Whether there exists a situation where “the victim is uglier than the murderer”? (A meaningless intellectual game)
- Whether the victims “wanted to die”? (Freudian absurd speculation — the so-called “death wish”)

Arendt points out that the book deals only with a “sadly limited” subject — **a report on the trial of one specific person (Eichmann)**. It is not a general history of the Holocaust, not a theoretical analysis of totalitarianism, and not a philosophical treatise on the nature of evil.

60.1.2. 2. The Strictly Factual Interpretation of the “Banality of Evil”

“When I speak of the banality of evil, I do so only on the strictly factual level, pointing to a phenomenon which stared one in the face at the trial.”

This is the book’s most crucial passage:

- Eichmann was **not** Iago or Macbeth — he was not a dramatically villainous figure filled with demonic motives. Nor had he “decided to prove himself a villain” (like Richard III).
- Apart from an “extraordinary diligence” regarding personal advancement, he had **no motives whatsoever** — and diligence alone is not criminal.
- He “merely, to put the matter colloquially, **never realized what he was doing**.” This “lack of imagination” enabled him to sit across from a German Jewish police interrogator and pour out his heart — about why he had only reached the rank of Lieutenant Colonel — without ever grasping the absurdity and horror of the situation.
- “He was not stupid. It was **sheer thoughtlessness** — something by no means identical with stupidity — that predisposed him to become one of the greatest criminals of that period.”

Core judgment: “**This remoteness from reality and this thoughtlessness can wreak more havoc than all the evil instincts taken together which are perhaps inherent in man — this, in fact, was the lesson one could learn in Jerusalem.**”

60.1.3. 3. The Concept of “Administrative Massacres”

Arendt notes that while the concept of “genocide” applies to this case, it is **not precise enough** — because “massacres of whole peoples are not without precedent in history.” She proposes the term “administrative massacres” to capture the **distinctiveness** of Nazi crimes:

- Such massacres can be directed against **any arbitrary group** — the criterion of selection depends only on circumstances. Hitler began with “euthanasia” (targeting the “incurably ill”) and planned to end by eliminating “genetically damaged Germans” (heart and lung patients).
- “In the not too distant future of an automated economy, people might be tempted to eliminate all those whose IQ falls below a certain level.” — This is a stunningly prophetic Arendtian insight.
- The key feature is: this is a large-scale program of elimination organized by the state apparatus through **administrative procedures** (not war, not passion).

60.1.4. 4. Philosophical Reflections on Justice and Trial

In the Postscript, Arendt articulates her understanding of the trial itself:

- The central concern of a trial must be **the defendant as a person** — a flesh-and-blood individual with a personal history, not a symbol of an “epoch” or a “people”
- She rejects both the attempt to turn Eichmann into a “scapegoat” and the attempt to “find Eichmann in each of us” — both approaches dissolve the meaning of a concrete trial
- She cites Grotius’s ancient insight about the meaning of punishment: punishment is necessary “to defend the honor or authority of him who was hurt, lest the failure to punish cause his position to sink”

60.2. Reading Guidance

The Postscript is Arendt’s calm response after *Eichmann in Jerusalem* provoked a massive public controversy. It condenses the book’s central claims:

- 1) **Evil can be “banal”** — not demonic, but thoughtless and bureaucratic

- 2) **Thoughtlessness is extraordinarily dangerous** — it can wreak more havoc than any evil instincts
- 3) **Justice must be concrete** — directed at individuals, not at epochs, peoples, or human nature

61. The Contemporary Relevance of Arendt's Thought

Hannah Arendt's thought has extraordinarily urgent relevance in our era of global political polarization, internet-driven atomized society, and algorithmic authoritarianism:

- **Vigilance against the absorption of politics by technology and consumerism:** She presciently warned that modern society is transforming into a purely “consumer society.” People devote most of their energy to consumption for survival (Labor) and technical efficiency (Work), while withdrawing from concern for public affairs (abandoning Action). This political indifference is precisely the breeding ground for the resurgence of totalitarianism or populism.
- **Reaffirming the importance of independent thinking:** In the face of information torrents and group polarization, the risk of the “banality of evil” still exists. Arendt calls on everyone to maintain critical thinking in the public space, not blindly obeying any absolute ideology.
- **Defending “public life”:** In an age when social media isolates people in their respective “information cocoons,” Arendt reminds us that we must rebuild genuine “public space,” listen face-to-face to the speech of those different from ourselves, and act together on the premise of embracing human “plurality.”

W. V. Quine

Notes compiled from the instructor's lecture materials and reading; for knowledge sharing only; not for any commercial use

62. Background

Willard Van Orman Quine (1908–2000): one of the most influential American philosophers of the 20th century, a central figure in the analytic philosophical tradition. His work spans logic, the philosophy of language, epistemology, metaphysics, and the philosophy of science.

Quine was deeply influenced by Russell and Carnap, but ultimately arrived at a fundamental critique of Logical Empiricism. His philosophy can be understood as a kind of **naturalized epistemology** — pulling epistemology down from the status of “first philosophy” and making it part of empirical psychology.

Major works:

- 1951: “Two Dogmas of Empiricism”
- 1953: *From a Logical Point of View*
- 1960: *Word and Object*
- 1995: *From Stimulus to Science*

63. Two Dogmas of Empiricism

Quine’s “Two Dogmas of Empiricism” (1951) is one of the most revolutionary papers in the history of analytic philosophy. He argues that traditional empiricism rests on two unproven dogmas:

63.1. Dogma 1: The Analytic-Synthetic Distinction

Since Hume and Kant, philosophers have distinguished two kinds of truths:

- **Analytic truths:** true by virtue of meaning alone, independent of fact (e.g., “All bachelors are unmarried”)
- **Synthetic truths:** true by virtue of fact (e.g., “Paris is in France”)

Quine argues: **The strict analytic-synthetic distinction cannot be maintained.** So-called “analytic truths” either rely on the equally obscure concept of synonymy, or must ultimately appeal to empirical facts. There is no such thing as a purely non-empirical “truth of meaning.”

The core steps of this argument:

- 1) The concept of “analyticity” needs the concept of “synonymy” for its explanation
- 2) The concept of “synonymy” is explained either through definition (but definitions themselves rely on prior synonymy judgments), or through interchangeability (but this requires modal concepts, which are themselves problematic)
- 3) Therefore, the analytic-synthetic distinction is a “dogma of empiricism” — a metaphysical article of faith, not an empirically verifiable distinction

63.2. Dogma 2: Reductionism

Reductionism holds that every meaningful statement can be reduced to statements about immediate experience (such as the language of sense-data), and that each statement can independently be subjected to empirical verification or falsification.

Quine’s critique: **Our knowledge does not consist of individual statements facing the tribunal of experience one by one, but of the entire system of knowledge facing experience as a whole.** This is his “holism” or “confirmation holism.”

64. Naturalized Epistemology

64.1. From “First Philosophy” to Empirical Psychology

Traditional epistemology (since Descartes) conceived of itself as standing “before” or “above” science, providing a philosophical foundation for scientific knowledge. Quine overturns this picture:

“Epistemology, or something like it, simply falls into place as a chapter of psychology and hence of natural science. It studies a natural phenomenon, viz., a physical human subject. (Quine, “Epistemology Naturalized”, 1969)”

Epistemology is not the “foundation” of science, but part of science. We study how humans construct scientific theories from sensory stimuli (stimulus), and this study is itself an enterprise of empirical science.

In *From Stimulus to Science*, Quine traces the cognitive process from the most primitive sensory stimuli (“surface irritations”) to complex scientific theories. This is a natural history from physical input to theoretical output.

64.2. The Meaning of Naturalism

Quine’s naturalism involves several key commitments:

- There is no “first philosophy” — no philosophy that precedes science and provides its foundation
- The continuity of philosophy and science — there is no fundamental methodological difference between philosophical and scientific questions
- A re-understanding of traditional epistemological questions — “How do we know?” is no longer a normative question about justification, but a descriptive question about causal mechanisms

65. The Indeterminacy of Translation

65.1. The Thought Experiment of Radical Translation

Quine’s indeterminacy of translation thesis is the core of his philosophy of language. Imagine a linguist confronting a previously unencountered language (the “radical translation” scenario); she must construct a translation manual based on the natives’ verbal behavior.

Quine’s argument: **Different translation manuals can be compatible with all possible evidence of verbal behavior, yet be mutually incompatible.** That is, translation is not uniquely determined by behavioral evidence.

The classic “Gavagai” example: the native says “Gavagai” when a rabbit runs by. Should this word be translated as:

- “Rabbit”?
- “Undetached rabbit part”?
- “Temporal rabbit stage”?
- “Rabbithood” — the universal of being a rabbit?

All these translations are compatible with all observable verbal behavior. When you point and ask “Is this the same gavagai?” — no matter how you point, you are indicating the same rabbit, the same rabbit part, the same instance of a rabbit stage.

65.2. Philosophical Consequences

The deeper implications of the indeterminacy of translation:

- **Meaning is not an independent entity:** There is no “propositional meaning” independent of a translation manual waiting to be discovered
- **Inscrutability of reference:** Even in our own language, there is no fact of the matter determining what our words “really” refer to
- **Ontological relativity:** Questions about “what exists” can only be answered using a background language; there is no absolute answer

66. Ontological Commitment

Quine's slogan: "To be is to be the value of a variable."

This means:

- We should not ask empty metaphysical questions like "Does X exist?"
- Instead, we should ask: In order for our best scientific theories to be true, what entities must our theories quantify over?
- **Ontological commitment is revealed through quantificational structure**, not through a separate "existence" predicate

Quine's own ontology is austere: he admits the existence of physical objects and sets (classes), but rejects "intensional entities" such as propositions, meanings (as entities), properties, and possible worlds.

67. Critique of Modal Logic and Essentialism

Quine is a famous critic of modal logic. He holds that:

- Modal contexts ("necessarily...", "possibly...") are referentially opaque — they undermine the principle of substitutivity of identity
- The concept of "essential properties" is meaningless — what properties an object possesses "necessarily" depends on how we describe it, not on what the object itself essentially is
- Therefore, quantified modal logic is built on confusion

This position puts him in sharp opposition to Kripke (see the Kripke notes).

68. Quine's Relation to the Western Modern Philosophical Tradition

Quine's philosophy can be seen as a critical advance on the Frege-Russell-Wittgenstein tradition:

Scholar	Line of Inheritance	Core Divergence
Frege	Inherits the methodological tradition of logical analysis	Quine rejects Frege's view of "Thoughts" (Gedanke) as abstract entities, holding that meaning-entities are philosophically suspect
Russell	Inherits logical constructionism and the spirit of "Ockham's Razor"	Quine does not accept Russell's concept of "acquaintance" and sense-data epistemology, holding that all knowledge is ultimately theory-constructed
Carnap	Early deep influence from logical empiricism	Quine's "Two Dogmas" directly destroyed Carnap's analytic-synthetic distinction and reductionist program
Wittgenstein	Inherits the deep concern with language and meaning	Quine does not endorse the Wittgensteinian "therapeutic" philosophical method or the view of "meaning as use"; he leans more toward scientific systematic theory construction

69. Assigned Reading Analysis: From Stimulus to Science (1995), Chapter II: Naturalism, pp. 15–26

The assigned pages (15–26) cover Chapter II, "Naturalism," of Quine's late work — the most mature statement of his "naturalized epistemology" program.

69.1. The Structure and Starting Point of Chapter II

In this chapter, Quine traces the natural history from the most primitive sensory stimuli to scientific theory. His core question is: how do we begin from “surface irritations” of the retina and eventually construct complex scientific theories about quarks, black holes, and DNA?

The key advance of Chapter II is: **Quine here is not merely negatively critiquing traditional epistemology, but positively describing his own alternative.**

69.2. From Stimulus to Observation Sentences

Quine’s naturalized epistemology begins with the most “basic” level: **observation sentences**. Observation sentences have two features:

- 1) They are “occasion sentences” — the speaker can judge their truth or falsity on the spot by checking sensory stimuli
- 2) They are “intersubjectively agreed” — all members of the same linguistic community, under the same stimulus conditions, will agree on the truth-value of the sentence

“It is raining,” “This is red” — these are paradigm observation sentences. They lie at the outermost layer of the “web of science,” directly facing the tribunal of experience.

Note: Even at the level of observation sentences, **sociality and publicity have already entered** — observation sentences are not about private sensations, but about publicly verifiable states of affairs.

69.3. The Evolution of Theoretical Language

Quine describes a gradual progression from observation sentences to increasingly abstract theoretical sentences:

- Observation sentences: respond directly to sensory stimuli
- Observation categoricals: classify observation sentences (“this dog,” “that cat”)
- Theoretical entities: electrons, genes, black holes — these cannot be directly observed, but are indirectly linked to observation sentences through the **hypothetico-deductive method**
- Mathematics and logic: lie at the innermost layer of the scientific web — farthest from direct experience, but also hardest to unsettle by empirical refutation

69.4. “All Science Is a Human Construction” (p. 19)

Quine’s core naturalistic thesis is condensed in a key statement in Chapter II:

Our entire scientific theory — from the most everyday common sense to the most abstract theoretical physics — is merely a conceptual apparatus used by humans to predict and control future sensory stimuli. There is no “God’s eye view,” no external point of comparison between “the real world” and “theory” that is independent of that apparatus.

But this does not mean Quine is an “anti-realist” or a “relativist.” He holds that **science is our best — our only — way of knowing the world.** The point of “naturalized epistemology” is not to deny the cognitive authority of science, but to point out: science does not need to seek an “a priori” or “metaphysical” foundation for itself — it can study itself, how it grows from stimuli.

69.5. The Final Rejection of “First Philosophy” (pp. 22–6)

In the latter part of Chapter II, Quine explicitly rejects the “first philosophy” tradition stemming from Descartes. Traditional philosophy conceives of itself as the foundation of science — philosophy’s task is first to establish certain indubitable principles, upon which science’s legitimacy can then be built.

Quine’s response:

- 1) This is impossible — epistemology itself must make use of scientific knowledge (e.g., psychology, linguistics, evolutionary theory)

- 2) This is unnecessary — science can improve itself through **self-correction**; it does not need an “external” philosophical judge
- 3) The mission of philosophy is redefined: philosophy is not the foundation of science, but its extension — it is part of “natural science in the broadest sense”

69.6. Reading Guidance

Chapter II is the best entry point for understanding Quine’s mature philosophy. Unlike “Two Dogmas,” it is not negatively destructive, but positively sketches a complete picture of “naturalized epistemology” from sensory stimuli to science. Its core message is twofold: (1) Epistemology is possible — but only as part of empirical science; (2) Philosophy has not disappeared, but has been repositioned — it is no longer the king but a citizen, no longer standing above science, but walking alongside it.

70. Summary: Quine’s Philosophical Legacy

Quine’s thought can be condensed into several core theses:

- 1) **Rejection of the analytic-synthetic distinction:** There is no purely “truth of meaning” — all knowledge is ultimately continuous with experience
- 2) **Confirmation holism:** Our belief system confronts experience as a whole — any individual statement can be retained in the face of counterevidence by adjusting other parts of the system
- 3) **Naturalized epistemology:** Epistemology is a chapter of psychology — we study how humans construct science from sensory stimuli
- 4) **Indeterminacy of translation:** Meaning is not a fact independent of a translation manual
- 5) **Criterion of ontological commitment:** “To be is to be the value of a variable” — ontological commitment is revealed through quantificational structure

“Our talk of external things, our very notion of things, is just a conceptual apparatus that helps us to foresee and control the triggering of our sensory receptors in the light of previous triggering of our sensory receptors.” — W. V. Quine, *From Stimulus to Science*

Alan Turing

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71. Background

Alan Turing (1912–1954): British mathematician, logician, and computer scientist, acclaimed as “AI’s first philosopher.” His work spans mathematics, logic, cryptanalysis, and philosophy, laying the theoretical foundations of modern computer science and artificial intelligence.

Major contributions:

- 1936: “On Computable Numbers,” proposing the concept of the universal Turing machine
- 1939–45: Code-breaking work during World War II
- 1950: “Computing Machinery and Intelligence,” proposing the famous Turing Test

Turing was an exemplary scholar, especially as AI’s first philosopher. Vision, drive, interdisciplinarity, integrity.

72. Assigned Reading Analysis: Computing Machinery and Intelligence (1950), pp. 433–4, 442–54

The assigned reading covers two crucial parts of this classic paper: the opening set-up of the “Imitation Game” (pp. 433–4) and the later responses to various objections (pp. 442–54).

72.1. pp. 433–4: Setting Up the Imitation Game

72.1.1. Substituting the Question Rather Than Answering It

Turing opens by announcing: “I propose to consider the question, ‘Can machines think?’” But he immediately points out that attempting to answer this by defining “machine” and “think” is “dangerous” — if defined according to the common use of these words, the answer would reduce to a statistical result like a “Gallup poll.” Therefore, **he does not answer the question with definitions — he replaces it with another question.**

72.1.2. The Rules of the Imitation Game

The original imitation game involves three people: a man (A), a woman (B), and an interrogator (C). The interrogator communicates with the other two by typewritten text, aiming to determine which is the man and which is the woman. Turing’s proposed new form is: **“What will happen when a machine takes the part of A in this game?”** If, after adequate interaction, the interrogator cannot reliably distinguish the machine from the human, then the machine has passed the test.

Turing particularly emphasizes: the design of the test isolates “physical and intellectual capacities.” The interrogator cannot see, touch, or hear the participants — only typewritten text. This focuses the question on **intelligence**, rather than on mimicking appearance or voice.

72.1.3. Sample Q&A

Turing provides several sample Q&As to show the range of the test:

- Q: “Please write me a sonnet on the subject of the Forth Bridge.” A: “Count me out on this one. I never could write poetry.”
- Q: “Add 34957 to 70764.” A: “(Pause about 30 seconds and then give as answer) 105621.”
- Q: “Do you play chess?” A: “Yes.” (Followed by a chess problem with K and R against K)

These examples demonstrate that the test can cover any domain of human activity — from arithmetical reasoning to humor, from games to literature — without penalizing the machine for lacking certain non-intellectual human traits (such as lacking charm in a beauty contest).

72.2. pp. 442–4: Responding to Various Objections

In the latter portion of the paper (pp. 442–54), Turing systematically responds to nine objections to machine intelligence. The assigned pages cover the most critical ones:

72.2.1. The Mathematical Objection (pp. 444–5)

This is the most important philosophical objection — based on Gödel’s incompleteness theorems:

“There are certain things that [a machine] cannot do. If it is rigged up to give answers to questions, there will be some questions to which it will either give a wrong answer, or fail to give one at all.”

Turing’s response proceeds in layers:

- 1) **Humans also make mistakes:** “We too often give wrong answers to questions ourselves to be justified in being very pleased at such evidence of fallibility on the part of the machines.”
- 2) **Superiority is relative:** Our feeling of superiority can only be a “petty triumph” over that one particular machine — it is impossible to triumph simultaneously over all machines. “There might be men cleverer than any given machine, but then again there might be other machines cleverer again, and so on.”
- 3) **Machine learning:** Using machine learning methods, machines can also “master or prove the relevant facts” — this is no longer about fixed formal systems.

72.2.2. The Argument from Consciousness (pp. 445–6)

“Not until a machine can write a sonnet or compose a concerto because of thoughts and emotions felt, and not by the chance fall of symbols, could we agree that machine equals brain — that is, not only write it but **know that it had written it.**”

Turing’s response: this is a solipsistic objection. “A may think ‘A thinks, while B is merely pretending’ — and yet, what is the criterion for not pretending?” Taking this objection to its extreme logical conclusion, the only way to know that a machine is thinking is to be that machine and feel oneself thinking. This would make communication about other minds completely impossible.

Turing’s humorous reply: instead of endless argument, “it would be better to accept the polite convention that everyone thinks.”

72.2.3. On the Criterion of “Genuine Thinking”

Throughout the paper, Turing consistently opposes treating “thinking” as some mysterious essence or unverifiable inner state. His philosophical strategy is **operationalist** — not defining what thinking “is,” but providing an operationalizable test to assess it. But this does not mean he is a “behaviorist” — he merely holds that if we ultimately cannot distinguish “genuine thinking” from “mere imitation” in an operational way, then perhaps the distinction itself is not as clear and useful as we imagine.

72.3. Reading Guidance

The enduring influence of this paper lies in its success in transforming a profound philosophical question (“What is thought?”) into a technical, discussable form — while maintaining deep philosophical sensitivity to the question. Turing does not simply evade the metaphysical question, but handles it with an engineer’s precision and a philosopher’s prudence.

73. The Turing Test (The Imitation Game)

73.1. Core Problem and Method

In his 1950 paper, Turing posed a fundamental question: “Can machines think?” But he did not directly answer this question; instead, he replaced it with a behavioral test.

“I propose to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’... Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.”

Turing’s proposed “Imitation Game” is: an interrogator (C) communicates by typewritten text with two participants in separate rooms (A and B), where A is a machine and B is a human. If, after adequate interaction, the interrogator cannot reliably distinguish the machine from the human, then the machine has passed the test.

73.2. Common Misunderstandings of the Turing Test

Many textbooks misrepresent the Turing Test:

- Misunderstanding 1: “Turing rejected the question ‘Can machines think?’” — In fact, Turing merely replaced it with a clearer behavioral test, not rejecting the question itself
- Misunderstanding 2: “Passing the test is equivalent to thinking” — Turing himself did not hold that passing the test is a sufficient condition for thinking; he treated the test as a conceptual tool, a starting point for thinking rather than an endpoint for answering
- Misunderstanding 3: “The Turing Test is behavioristic” — Searle and others have criticized it this way, but Turing’s intentions are far subtler than behaviorism

Turing himself said in a 1952 BBC broadcast: “You might call it a test to see whether the machine thinks, but it would be better to avoid begging the question and say that the machines that pass are (say) ‘Grade A’ machines... My suggestion is merely that this is what we ought to discuss. It is not the same as ‘Can machines think?’ but it seems close enough for our present purposes, and raises roughly the same difficulties.”

73.3. The Design Philosophy of the Test

The Turing Test is designed with careful deliberation:

- Excluding the interference of physical appearance through typewritten communication (“no engineer or chemist claims to be able to produce a material which is indistinguishable from the human skin”)
- Allowing the machine to display various human abilities (mathematics, chess, poetry composition, etc.)
- Not penalizing the machine for lacking certain non-intellectual human traits (such as enjoying good food)

Turing humorously comments: “Possibly a machine might be made to enjoy this delicious dish [strawberries and cream], but any attempt to make one do so would be idiotic.”

74. The Mathematical Objection and Gödel’s Theorem

74.1. The Background of Gödel’s Incompleteness Theorems

Gödel’s Theorem: No axiomatization, that is, no finite collection of basic principles and rules (i.e., no algorithm), can ever be strong enough to establish all and only the truths of arithmetic.

Gödel himself wrote: “The human mind is incapable of formulating ... all its mathematical intuitions, i.e., if it has succeeded in formulating some of them, this very fact yields new intuitive knowledge ... This fact may be called the ‘incompleteness’ of mathematics.”

74.2. The Lucas-Penrose Argument (The Mathematical Objection)

Lucas argued in 1961: because of Gödel’s incompleteness theorems, any machine (as a formal system) has true propositions it cannot prove, but the human mind can “see” that these propositions are true. Therefore, the human mind transcends any machine.

“However complicated a machine we construct, it will ... correspond to a formal system, which in turn will be liable to the Gödel procedure of finding a formula unprovable-in-that-system. This formula the machine will be unable to produce as being true, although a mind can see that it is true. ... Thanks to Gödel’s theorem, the mind always has the last word.”

74.3. Turing's Response

Turing anticipated this objection in 1950 and responded on two levels:

First response: Humans themselves make mistakes. "We too often give wrong answers to questions ourselves to be justified in being very pleased at such evidence of fallibility on the part of the machines."

Second response: Human superiority can only be demonstrated against one particular machine; it cannot simultaneously defeat all machines. "There would be no question of triumphing simultaneously over all machines. In short, then, there might be men cleverer than any given machine, but then again there might be other machines cleverer again, and so on."

Other responses:

- Some axiomatic systems are also too complex for humans to grasp
- Machine learning can enable machines to learn and prove relevant facts just as humans do

75. Searle's Chinese Room Argument

75.1. The Structure of the Argument

Searle's "Chinese Room" thought experiment seeks to refute "Strong AI" — the view that "an appropriately programmed computer really is a mind."

Searle imagines: a person who does not understand Chinese is shut in a room, processing Chinese characters according to an English rulebook. From the outside, they seem to "understand" Chinese, but in fact they understand nothing. Searle holds that a computer is like this person — merely manipulating symbols, without genuine understanding.

Key premise: "In the Chinese case I have everything that artificial intelligence can put into me by way of a program, and I understand nothing."

75.2. Criticisms of Searle's Argument

- **Combination reply:** If Searle were placed inside the brain of a real speaker and he still did not understand, does this mean his test itself is flawed?
- **Searle** claims that attributing mental states to something "not made of the same stuff as us" is unnatural and mistaken
- The Chinese Room argument does not truly support Searle's conclusion — it exposes the need for a better experiment

76. The Imitation World Argument

76.1. From the Turing Test to the Imitation World

Developments in modern AI (such as ChatGPT) have called the sufficiency of the traditional Turing Test into question. The course proposes an upgraded philosophical test:

Imitation World Argument:

- 1) P1: In an alternative history, machines would possess "human mindedness"
- 2) P2: Our human mindedness = "human mindedness" in the alternative history
- 3) C: Machines can acquire human mindedness

This is a philosophical deepening of the traditional Turing Test, upgrading from "can they deceive humans in conversation" to "can they become human in a complete social history."

76.2. The Argument from Robot Existentialism

The counter-argument holds:

- 1) P1: In the alternative history, machines would be inauthentic
- 2) P2: Our human mindedness $\neq$ "human mindedness" in the alternative history
- 3) C: Machines cannot acquire human mindedness

77. Ten Objections Surrounding the Imitation World

- (1) **The Biological Objection:** Opposes Strong AI, arguing that biology is necessary for mindedness
- (2) **Unrealistic:** This alternative history is completely unrealistic
- (3) **Epistemically Implausible:** Ignorance requires absurdity to sustain
- (4) **Verbal Dispute:** This is merely a dispute about word definitions
- (5) **Gödelian Objection:** A continuation of the Lucas-Penrose argument
- (6) **Functionalist Presupposition:** The argument presupposes multiple realizability
- (7) **Semantic Externalism:** Different worlds imply different concepts
- (8) **Phenomenological Objection:** The imitation world is inconceivable
- (9) **Searle's Objection:** Don't take the epistemological route
- (10) **Conceptual Limitations:** Our concepts may be unable to grasp important differences

78. Turing's Intellectual Legacy

78.1. The Relationship Between Turing, Gödel, and the Philosophy of AI

Turing's thought, together with Gödel's incompleteness theorems, constitutes a philosophical framework concerning mind and machine:

- **Gödel** revealed the boundaries of formalized rationality — any closed system cannot encompass the totality of truth
- **Turing** posed a more pragmatic question: rather than arguing whether machines “can” think, ask to what extent they can exhibit intelligent behavior
- **Modern developments:** Large language models (like ChatGPT) rely neither on fixed axiomatic systems nor are they classical symbolic AI — this shifts the form of the challenge posed by Gödel's theorem; the question is no longer “can formal systems simulate humans,” but “can statistical machines produce genuine understanding”

78.2. Key Philosophical Insights

Turing's deepest contribution lies not in providing a definitive answer to “Can machines think?”, but in:

- 1) Transforming a philosophical question into an operational empirical test
- 2) Recognizing that intelligence is not an “all-or-nothing” property, but a spectrum of capacities with degrees
- 3) Foreseeing the importance of machine learning and adaptive systems
- 4) Transcending the simple dichotomy between behaviorism and naïve mentalism

“The original question, ‘Can machines think?’ I believe to be too meaningless to deserve discussion. Nevertheless I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted.”
— Alan Turing, 1950

Iris Murdoch

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79. Background

Iris Murdoch (1919–1999): Irish-born British philosopher and novelist. Her philosophy integrates the analytic and continental traditions, influenced by Wittgenstein, Heidegger, Sartre, and Simone Weil. She published 26 novels, winning the Booker Prize in 1978.

1938–42 Oxford University (Somerville College), classmates with Elizabeth Anscombe, Philippa Foot, and Mary Midgley. 1948–63 taught at Oxford. She swam against the prevailing behaviorist current at Oxford, insisting on the centrality of inner moral life and defending moral realism under the banner of Plato.

Major philosophical works:

- 1953/87: *Sartre: Romantic Rationalist*
- 1970: *The Sovereignty of Good*
- 1977: *The Fire and the Sun: Why Plato Banished the Artists*
- 1992: *Metaphysics as a Guide to Morals*

80. Core Thought: Returning to the Inner

80.1. Critique of Modern Moral Philosophy

Murdoch holds that modern moral philosophy (behaviorism, existentialism, utilitarianism) jointly constructs an image of “modern man,” but this image is impoverished:

“I find the image of man which I have sketched above both alien and implausible. That is, more precisely: I have simple empirical objections (I do not think people are necessarily or essentially ‘like that’), I have philosophical objections (I do not find the arguments convincing), and I have moral objections (I do not think people ought to picture themselves in this way).”

The features of this image of “modern man”:

- **Behaviorism:** Equating the existence of meaning and action with what is publicly observable
- **Existentialism:** Eliminating the substantive self, emphasizing the lonely and omnipotent choosing will
- **Utilitarianism:** Assuming that morality concerns (and can only concern) public actions

In this picture, morality is reduced to public, visible **behavioral choices** and **decisions of the will**, while inner moral life — the quality of attention, the way of seeing, the capacity to love — is entirely neglected.

80.2. Recovery of the Inner

Murdoch's core claim: **Moral philosophy must take the inner life of persons seriously.** We are not merely agents, but also perceivers, attenders, imaginers.

She illustrates this with a vivid example:

M is a mother who feels hostility toward her daughter-in-law, D. M finds D to be a good-hearted girl, but not exactly common yet certainly unpolished and lacking in dignity and refinement — in short, her son has married “beneath him.” But M behaves with perfect propriety, never letting her true feelings show. M engages in an inner struggle; she may be tempted to enjoy caricatures of D in her imagination...

“M looks at D, she attends to D, she focuses her attention. M is engaged in an internal struggle. She may for instance be tempted to enjoy caricatures of D in her imagination. ... This activity, as I said, could be described in a variety of ways, but one very natural way is by the use of specialized normative words, what one might call the secondary moral words in contrast to the primary and general ones such as “good”. M stops seeing D as “bumptious” and sees her as “gay”, etc.”

The key point: moral change occurs in a **shift in the inner way of seeing** — not through a public behavioral choice, but through patient, loving attention that re-sees another person.

81. Moral Realism and the Concept of “The Good”

81.1. “The Good” Is an Objective Reality

Murdoch inherits G.E. Moore’s position: **“Good” is an objective, indefinable reality.** She opposes modern philosophy’s reduction of “good” to a function of the will or a projection of emotion.

“I have spoken of efforts of attention directed upon individuals and of obedience to reality as an exercise of love, and have suggested that ‘reality’ and ‘individual’ present themselves to us in moral contexts as ideal end-points … Here we retrieve the deep sense of the indefinability of good … Good is indefinable … because of the infinite difficulty of the task of apprehending a magnetic but inexhaustible reality.”

The Good is not an object we can fully grasp; it is a “magnetic but inexhaustible reality” toward which we can only asymptotically approach through sustained moral effort.

81.2. “Unselfing”

Murdoch holds that our “fat, relentless ego” constantly interferes with our moral attention to others and the world. Therefore, moral progress requires a process of “unselfing.”

Beauty is an important instrument of “unselfing”:

“I am looking out of my window in an anxious and resentful state of mind, oblivious of my surroundings, brooding perhaps on some damage done to my prestige. Then suddenly I observe a hovering kestrel. In a moment everything is altered. The brooding self with its hurt vanity has disappeared. There is nothing now but kestrel. And when I return to thinking of the other matter it seems less important…”

In the experience of beauty, the self temporarily vanishes; we purely attend to something that is not ourselves. This is precisely the model of moral attention: selflessly, lovingly directed toward reality.

Murdoch credits this concept to Simone Weil’s notion of “décreation.”

82. Moral Language: Thick and Thin Concepts

82.1. Primary vs Secondary Moral Words

Murdoch distinguishes two kinds of moral vocabulary:

- **Primary moral words:** such as “good,” “bad” — abstract, universal, thin in content
- **Secondary moral words:** such as “generous,” “tactful,” “courageous,” “bumptious,” “gay” — concrete, rich, simultaneously descriptive and evaluative

This was later developed by Bernard Williams into the distinction between “thin concepts” and “thick concepts.” However, Williams and modern textbooks almost never acknowledge Murdoch’s priority in this contribution.

The importance of thick concepts: they demonstrate that moral language is not merely the expression of personal attitudes or the issuance of commands — the words themselves carry nuanced perception and understanding of the world.

82.2. The Contingent Privacy of Language

Murdoch puts forward a radical and profound view: **Moral language is inevitably situational, private, and difficult to access.**

“This dependence of language upon contexts of attention has consequences. Language is far more idiosyncratic than has been admitted. Reasons are not necessarily and qua reasons public. They may be reasons for a very few, and none the worse for that. “I can’t explain. You’d have to know her.” … Moral language which relates to a reality infinitely more complex and various than that of science is often unavoidably idiosyncratic and inaccessible.”

This means:

- Moral understanding, unlike scientific knowledge, cannot be fully made public
- “Knowing the meaning of a word” has two senses: one linked to ordinary public language, the other deeper — knowledge of value concepts requires “deep” understanding
- The process of understanding is “forward, in the direction of the more and more private,” not “backward, tracing to some impersonal public language rules”

83. Against the Behaviorist-Existentialist-Utilitarian Trinity

83.1. Dismantling the Idol of “Will”

Murdoch points out that modern moral philosophy deifies “will” and “choice.” Existentialism (especially Sartre) reduces morality entirely to the lonely, arbitrary decision of the will — as if, when facing a moral dilemma, a person has only an empty, isolated “moment of choice.”

But Murdoch holds that **the real moral work happens before choice** — in everyday attention, imagination, and reflection, we gradually shape the way we see the world. So-called “choice” is merely the tip of an iceberg of long-term inner work.

“The idea that ‘good’ is a function of the will stunned philosophy with its attractiveness, since it solved so many problems at one blow: metaphysical entities were removed, and moral judgments were seen to be, not weird statements, but something much more comprehensible, such as persuasions or commands or rules.”

But the cost is immense — it erases the entire domain of inner moral life.

83.2. Critique of “Scientism”

Murdoch acutely identifies the root of the maladies of modern moral philosophy: “the domination of inexact ideas of science which haunt philosophers and other thinkers.”

“Psychoanalysis ... The notion of an ‘ideal analysis’ is a misleading one. There is no existing series the extension of which could lead to such an ideal. This is a moral question; and what is at stake here is the liberation of morality, and of philosophy as a study of human nature, from the domination of science: or rather from the domination of inexact ideas of science which haunt philosophers and other thinkers.”

84. The Wittgenstein-Heidegger-Murdoch Dialogue

Murdoch’s philosophy can be seen as a critical dialogue with Wittgenstein and Heidegger:

- **Engaging Wittgenstein:** Murdoch cites Wittgenstein on the language of sensation — “If we construe the grammar of the expression of sensation on the model of ‘object and name,’ the object drops out of consideration as irrelevant” (PI §293). She endorses Wittgenstein’s critique of referentialism, but holds that Wittgenstein is too focused on public language, neglecting the private depth of inner moral experience.
- **Engaging Heidegger:** Murdoch agrees with Heidegger’s critique of “das Man” and everyday fallenness — “one of the characteristics of morality is that one cannot entirely rest at the level of convention, and that in some ways one ought not to remain there.” But she emphasizes achieving authenticity more through **love and attention** (rather than solely through Being-towards-death).
- **Murdoch’s unique contribution:** Restoring Platonic moral realism within the analytic tradition — the Good is objective, but can only be asymptotically approached through sustained, loving attention.

85. Assigned Reading Analysis: The Sovereignty of Good (1970), “The Idea of Perfection”, pp. 1–9

The assigned pages (1–9) are the opening of “The Idea of Perfection,” the first essay in Murdoch’s *The Sovereignty of Good* — the core text of her moral philosophy.

85.1. The Paper's Starting Point: Philosophy's Two-Way Movement (p. 1)

Murdoch opens with philosophy's "return to the beginning" movement:

"Philosophy has in a sense to keep trying to return to the beginning: a thing which it is not at all easy to do. There is a two-way movement in philosophy, a movement towards the building of elaborate theories, and a move back again towards the consideration of simple and obvious facts. McTaggart says that time is unreal, Moore replies that he has just had his breakfast. Both these aspects of philosophy are necessary to it."

Murdoch positions herself at the "return movement" end — not constructing ever more complex theoretical edifices, but returning to those simple, obvious facts that contemporary moral philosophy has "theorized away." She particularly identifies two forgotten facts: "an unexamined life can be virtuous" and "**love** is a central concept in morality."

85.2. Critique of the "Modern Man" Image (pp. 2–8)

Murdoch develops her analysis through a critical engagement with Stuart Hampshire, sketching the image of "modern man" constructed by the contemporary mixture of behaviorism + existentialism + utilitarianism:

First, all meaning lies in publicly observable behavior. Hampshire maintains: "deliberation is qualifying only when expressed as audible speech or visible action"; "the inner world is unavoidably parasitic upon the outer world"; "nothing counts as an action unless it brings about a recognizable change in the world."

Second, the idolization of will and choice. In Hampshire's framework, "I identify with my will" — I am what I choose and do, not what I perceive and feel. Thoughts have nothing to do with me (they "go their own way"); only the will's choices express "me."

Third, the fantasy of the "ideal rational man." Hampshire portrays an "ideally rational man" — one who is conscious of all his memories, whose desires are all subordinated to definite future possibilities, who can fully distinguish present situations from unconscious past memories. This ideal person does not exist — and Murdoch asks: would we want him to exist? "Ideal rationality would deprive us of art, of dreams, of imagination, of likes and dislikes not tied to instinctual needs."

85.3. M and D: A Demonstrative Moral Narrative (pp. 16–17)

Murdoch introduces a concrete story to illuminate her own alternative moral picture:

"A mother, whom I shall call M, feels hostility to her daughter-in-law, whom I shall call D. M finds D quite a good-hearted girl, but while not exactly common yet certainly unpolished and lacking in dignity and refinement. ... M does not like D's accent or the way D dresses. M feels that her son has married beneath him."

Key design: M **never expresses** her true feelings — her behavior is consistently impeccable. This means that, in this case, **all the moral work happens entirely internally**. M engages in inner struggle through sustained, patient "attending" to D — "she may be tempted to imagine and caricature D." But she strives to "see D with just and loving attention."

Moral transformation occurs when M changes the way she sees D* — when she no longer sees D as "bumptious" but as "gay" — this transformation occurs not through a single public "choice," but through sustained, patient exercise of attention.

85.4. The Significance of Secondary Moral Words (pp. 22)

In the assigned pages, Murdoch develops her account of "secondary moral words." Words like "generous," "tactful," "courageous," "vulgar," "bumptious" are simultaneously evaluative and descriptive — they both report facts and carry value. The richness of this kind of vocabulary demonstrates: "morality is not an isolated decision of the will, but a **way of seeing** the world."

85.5. Reading Guidance

In these 9 pages, Murdoch accomplishes a delicate philosophical task: through a careful reading of Hampshire (as the spokesman for “modern man”), she reveals the intrinsic limitations of the behaviorist-existentialist-utilitarian mixture — its placing of all the weight of morality on the public, visible, describable-as-“bringing-about-a-change-in-the-world,” while **systematically neglecting the invisible but more fundamental domain of inner moral life.**

86. Summary: The Core of Murdoch’s Moral Philosophy

Murdoch’s philosophy can be summarized in the following core theses:

- 1) **The real field of morality is inner life:** Attention and ways of seeing are the core categories of morality, not merely public behavioral choices
- 2) **The Good is an objective reality:** The Good is not a projection of the will or a discharge of emotion, but a “magnetic but inexhaustible reality” toward which we approach through moral effort
- 3) **Language and moral understanding are situational:** Moral language is unavoidably, to some extent, private and difficult to access — in contrast with the public language of science
- 4) **Love is the core capacity of moral cognition:** Love is not the opposite or an appendage of morality, but the capacity to see others and the world as they truly are
- 5) **“Unselfing” is the key to moral progress:** The experience of beauty and loving attention can help us temporarily escape the tyranny of the self

“It is always a significant question to ask about any philosopher: what is he afraid of?” — Iris Murdoch, *The Sovereignty of Good*

Thomas Kuhn

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87. Background

Thomas S. Kuhn (1922–1996): American physicist, historian of science, and philosopher of science. His landmark work *The Structure of Scientific Revolutions* (1962/70) is one of the most cited works in the humanities and social sciences of the 20th century, fundamentally altering how people understand scientific development.

Life trajectory: 1943 Harvard B.S. in physics → 1949 Ph.D. in physics → turned to history of science → 1956 taught at Berkeley → 1964 Princeton → 1979 MIT

Core question: How does science develop? The traditional view holds that science progresses cumulatively — one fact added to another, gradually approaching truth. Kuhn's answer is: **No, scientific development alternates between “normal science” and “scientific revolutions.”**

88. Core Concepts

88.1. Paradigm

Paradigm is Kuhn's most central but also most perplexing concept. It has two layers of meaning:

Meaning 1 (broad): Disciplinary Matrix — the entire constellation of commitments shared by a scientific community, including:

- Symbolic generalizations (such as $F = ma$)
- Metaphysical commitments (such as “heat is molecular motion”)
- Shared values (such as the pursuit of simplicity, precision)
- Exemplary problems and their solutions

Meaning 2 (narrow): Exemplar — concrete problem-solutions in textbooks, the core instructional material through which scientists learn how to “do science.” Students learn to recognize and solve new problems by imitating exemplars.

“History, if viewed as a repository for more than anecdote or chronology, could produce a decisive transformation in the image of science by which we are now possessed. (Kuhn, *Structure*, p. 1)”

Kuhn's ambition is to change our fundamental understanding of science — not through philosophical reasoning, but through **historical facts**.

88.2. Normal Science

Normal science is research conducted under the guidance of an accepted paradigm. It has three features:

- 1) **Determinacy**: Scientists know what problems are important and what solutions are acceptable
- 2) **Depth not breadth**: Does not question foundational assumptions, but engages in refined “puzzle-solving” within the paradigm
- 3) **High consensus**: The scientific community has almost no dispute about basic theories, methods, and standards

The primary activity of normal science is **puzzle-solving**: fitting natural phenomena into the conceptual boxes provided by the paradigm. This stands in sharp contrast to Popper's “falsificationism” — in normal science, scientists do **not** attempt to falsify the paradigm, but attempt to demonstrate their ability to solve problems within the paradigm's framework.

88.3. Anomaly and Crisis

Anomalies are phenomena that normal science cannot digest — repeated failures of puzzle-solving within the paradigm's framework.

Conditions for anomalies to escalate into crisis:

- The anomaly involves the paradigm's foundational commitments
- The anomaly has significant practical application implications
- The anomaly has long resisted resolution by the most eminent scientists
- The number of anomalies is continuously increasing

During crisis, the scientific community begins to:

- Make ad hoc modifications to the existing paradigm
- Explicitly articulate and debate the paradigm's basic assumptions, which were never questioned during normal science
- Generate alternative theories (at this point entering "extraordinary science" or the "pre-paradigm" state)

88.4. Scientific Revolution

A scientific revolution is a **paradigm shift**. Kuhn likens it to a "Gestalt switch" — after the revolution, the very world the scientist inhabits has changed.

Classic cases of scientific revolutions:

Old Paradigm

Ptolemaic geocentrism

Aristotelian/impetus physics

Phlogiston theory

Caloric theory

Miasma / spontaneous generation

Action-at-a-distance electromagnetism

Blending inheritance / vitalism

Fixed-continent geology

Steady-state / eternal universe

New Paradigm

Copernican heliocentrism (1543–17th c.)

Newtonian mechanics (1660s–1687)

Lavoisier's oxidation theory (1770s–1789)

Thermodynamics / energy conservation (1820s–1850s)

Germ theory of disease (1850s–1880s)

Maxwell's electromagnetic field theory (1860s–1870s)

Mendelian genetics + Modern Synthesis (1900–1940s)

Plate tectonics (1960s)

Big Bang cosmology (1940s–1965)

88.5. Incommensurability

Paradigms before and after a revolution are "incommensurable," meaning:

- They employ different concepts and terminology (even the same word has different meanings — e.g., Newton's and Einstein's definitions of "mass")
- They pose different questions (what counts as a legitimate scientific question itself changes)
- They use different evaluative standards
- There is no paradigm-neutral standpoint from which to adjudicate between paradigms

But this does not mean that paradigm choice is "irrational" or "purely subjective." Kuhn emphasizes that the scientific community evaluates through shared values (accuracy, consistency, scope, simplicity, fruitfulness) — it is simply that these criteria are insufficient to uniquely determine choice.

89. Assigned Reading Analysis: The Structure of Scientific Revolutions (1962/70), Section 7: "Crisis and the Emergence of Scientific Theories"

The assigned section — Section 7, "Crisis and the Emergence of Scientific Theories" — is the turning point of the book. The first six sections establish the concepts of "paradigm" and "normal science"; from Section 7 onward, Kuhn turns to the explanation of how science undergoes revolution.

89.1. The Mechanism of Crisis Emergence

Kuhn's core thesis: **New theories emerge only after normal science has experienced persistent anomaly.**

An anomaly is different from a mere counterexample: only when the normal scientific community tries repeatedly and fails repeatedly — when the anomaly **persistently resists resolution** within the paradigm's framework — does the anomaly escalate into crisis.

Crisis has three features:

- 1) **The paradigm is “blurred”**: rules become flexible; scientists begin to experiment with more radical hypotheses that would not be entertained during normal science
- 2) **The emergence of meta-scientific discussion**: Foundational assumptions that were never questioned during normal science suddenly become subjects of open debate — scientists begin to do **philosophy** (about method, about metaphysics, about what counts as a legitimate explanation)
- 3) **Alternative theories begin to proliferate**: Different camps propose different new paradigms; the power of the old paradigm to unify science collapses

89.2. The Co-emergence of Crisis and New Theories

Kuhn points out that virtually all major scientific revolutions in history follow this pattern:

- Copernican astronomy emerged after centuries of accumulating anomalies in the Ptolemaic system
- Newtonian mechanics arose when Aristotelian physics faced countless “inexplicable” phenomena
- Lavoisier's chemical revolution occurred against the background of the phlogiston theory's repeated failures

89.3. “A Theory Is Declared Invalid Only When an Alternative Candidate Is Available” (p. 77)

This is one of Kuhn's most controversial claims and the most direct challenge to Popperian falsificationism. For Popper, when a theory is falsified, science should abandon it. But Kuhn points out: **Scientists never abandon a paradigm because it has been “falsified” — they only abandon it when there is a better alternative paradigm available.**

The reason: if they did not do this, science would descend into chaos — abandoning a “not quite perfect but workable” paradigm without any replacement is equivalent to abandoning “doing science” itself.

89.4. Crisis as a “Window of Opportunity”

Kuhn likens crisis to an “essential tension” — without crisis, there is no scientific revolution; but without normal science, crisis cannot occur (because only the deep puzzle-solving of normal science can reveal the true limitations of the paradigm). This is a paradoxical structure: **science's conservatism (refusing to abandon the paradigm) is precisely the condition for its revolutionary character (because only by holding on to the end can crisis be exposed).**

89.5. The Philosophical Significance of the Assigned Reading

Section 7 condenses several key tensions in Kuhn's philosophy of science:

- 1) **Science is rational, but it is not guided by fixed rules** — the “meta-scientific” arguments scientists make in crisis involve value judgments, not purely logical reasoning
- 2) **Progress does not require “toward truth”** — Kuhn's scientific progress is not an approach toward a God's-eye “real world,” but an improvement in puzzle-solving capacity
- 3) **Historical facts take priority over philosophical a priori** — Kuhn here is not engaging in a priori reasoning about “how science should be,” but observing what scientists actually did in the history of science

90. Scientific Progress and Truth

90.1. Can Progress Be Independent of “Truth”?

Kuhn raises a disturbing question: if paradigms constantly succeed each other and new paradigms are “incommensurable” with old ones, can we still say that science “approaches truth”?

Kuhn’s answer is **evolutionary epistemology**:

- The development of science is more like Darwinian evolution than linear progress toward a predetermined goal
- Later theories are “better” in terms of puzzle-solving ability, but this does not mean they are “closer to truth”
- “Truth” as a correspondence-with-reality concept loses clear meaning in Kuhn’s framework

“The developmental process described in this essay has been a process of evolution from primitive beginnings—a process whose successive stages are characterized by an increasingly detailed and refined understanding of nature. But nothing that has been or will be said makes it a process of evolution toward anything. (Kuhn, *Structure*, Postscript)”

90.2. The Problem of Relativism

Kuhn is often accused of being a relativist, but his position is more complex:

- He acknowledges that scientific development does produce better puzzle-solving tools
- But he denies that we can independently establish that “better puzzle-solving tools” equals “more accurate description of objective reality”
- Scientific knowledge is not a theory-independent “mirroring” of nature

91. Natural Science vs Social Science vs Philosophy

Kuhn’s object of analysis is the **natural sciences** (especially physics), but his theory has sparked intense discussion about the **social sciences**:

- Are the social sciences in a “pre-paradigm” stage? — Social scientists lack consensus on basic theories and methods, which is precisely the hallmark of “pre-paradigm” science
- But this does not mean the emergence of a “paradigm” in social science is merely a matter of time — the distinctive object of social science (conscious human agents) may make single-paradigm rule impossible

The role of philosophy: In Kuhn’s view, philosophy is not the “foundation” or “umpire” of science, but a reflective activity that becomes more important during periods of scientific crisis — when the metaphysical commitments of an old paradigm are being shaken.

92. The Influence and Controversies of Kuhn’s Thought

92.1. The Subversion of the Image of Science

Kuhn’s influence extends far beyond academic circles in the philosophy of science:

- Fundamentally changed science education — no longer presenting the history of science as the victor’s linear narrative
- Provided theoretical resources for the Sociology of Scientific Knowledge (SSK) and the “Strong Programme”
- Influenced all manner of “paradigm shift” discourse, from literary criticism to political theory

92.2. Major Controversies

- 1) **The vagueness of the paradigm concept**: Critics (especially Margaret Masterman) have pointed out that Kuhn uses “paradigm” in at least 21 different ways
- 2) **The charge of relativism**: If paradigms are incommensurable, are there rational standards for judging scientific progress?
- 3) **The debate with Popper**: Kuhn’s concept of normal science (scientists do not attempt to falsify the paradigm) forms a famous opposition to Popper’s critical rationalism
- 4) **Kuhn’s own revisions**: In the 1969 Postscript, Kuhn acknowledged the ambiguity of the “paradigm” concept and partially replaced it with “disciplinary matrix”

“Normal science does not aim at novelties of fact or theory and, when successful, finds none.” — Thomas S. Kuhn, *The Structure of Scientific Revolutions*

Saul Kripke

Notes compiled from assigned reading materials; for knowledge sharing only; not for any commercial use

93. Background

Saul A. Kripke (1940–2022): American philosopher and logician, one of the most original analytic philosophers of the 20th century. His *Naming and Necessity* (1972/1980) is one of the most far-reaching works in the history of analytic philosophy, utterly overturning the mainstream theory of names and reference that had held sway since Frege and Russell.

Kripke made foundational contributions to modal logic (possible worlds semantics) while still a teenager, but his greater philosophical impact came from his radical reconstruction of language, metaphysics, and epistemology.

Naming and Necessity originated as three lectures delivered at Princeton University in 1970, published as a book with minimal revision. The sheer originality of its content and the power of its arguments made it a classic of analytic philosophy.

94. The Theory of Names: Critique of Descriptivism

94.1. Descriptivism: The Frege-Russell Tradition

Since Frege and Russell, the mainstream view in analytic philosophy regarding proper names has been:

The meaning of a proper name is equivalent to a definite description or cluster of descriptions associated with it.

For example: the meaning of the name “Aristotle” might be “the greatest student of Plato,” “the teacher of Alexander the Great,” “the author of the *Nicomachean Ethics*,” etc.

This theory has many attractive features:

- It explains why “Aristotle = the author of the *Nicomachean Ethics*” is not trivially true (it provides information)
- It explains how names can refer to objects (through descriptions)
- It explains how we can talk about non-existent things (such as “Santa Claus,” “the golden mountain”)

94.2. Kripke’s Critique: The Modal Argument

Kripke’s core argument is: **if proper names are equivalent to descriptions, they should have the same modal properties — but they do not.**

Consider:

- 1) “Aristotle” (proper name)
- 2) “The teacher of Alexander the Great” (description)

If (1) and (2) have the same meaning, then:

- “If Aristotle exists, then Aristotle is the teacher of Alexander the Great” should be a necessary truth (analytic truth) — but it clearly is not! Aristotle could perfectly well have never taught Alexander.

Kripke’s conclusion: **Names are “rigid designators”; they designate the same object in every possible world.** Descriptions are not rigid — “the teacher of Alexander the Great” could designate different people (or no one at all) in different possible worlds.

94.3. Rigid Designator

Definition: A term is a rigid designator if and only if it designates the same object in every possible world in which that object exists.

- **Proper names** (“Aristotle,” “Paris”) are rigid designators
- **Natural kind terms** (“water,” “gold,” “tiger”) are also rigid designators

- **Definite descriptions** (“the teacher of Alexander the Great,” “the present King of France”) are typically **not** rigid designators

94.4. The Causal-Historical Theory of Reference

If a proper name does not determine its reference through a description, how does it do so?

Kripke’s alternative (sketched rather than fully developed) is: **A proper name is connected to the object it names through a “causal-historical chain.”**

- Initially, the object is assigned a name in some form of “initial baptism”
- Then, through a chain of communication (speakers passing on the referential intentions of others to the next person), the use of the name is propagated throughout the linguistic community
- As long as a speaker stands at the end of an appropriate historical causal chain, they can successfully use the name to refer to the original object — even if they possess no uniquely identifying description of that object

Important feature: This theory is not “you must know facts about the object” but rather “you must stand in an appropriate historical relation to the object.”

95. Necessity and Possible Worlds

95.1. Possible Worlds Semantics

Kripke’s conception of possible worlds differs from the extreme realism of David Lewis and others:

- **Possible worlds are not “distant planets”** — they are **ways the world could have been**
- Possible worlds are “stipulated,” not “discovered”
- When we say “Aristotle might not have taught Alexander,” we are not talking about some other planet where someone resembling Aristotle did not teach someone resembling Alexander — we are saying: this actual world might have been such that Aristotle did not become Alexander’s teacher

95.2. Necessary A Posteriori

The traditional distinction since Kant has been:

- Analytic = necessary = a priori
- Synthetic = contingent = a posteriori

Kripke breaks these equations. He argues for:

The existence of **Necessary a posteriori truths**:

- “Water is H₂O” — this truth was **discovered empirically** (a posteriori), but once discovered, we recognize: in any possible world, water is necessarily H₂O (necessarily water)
- “Hesperus is Phosphorus” — both are rigid designators (proper names), both designate the same object, so this identity statement is **necessary** — even though it required astronomical observation to know (a posteriori)
- “This table is made of this piece of wood” — if it is indeed made of this piece of wood, then in every possible world, it originates from this piece of wood (the essentiality of origin)

95.3. Contingent A Priori

Kripke also argues for the existence of contingent a priori truths:

- “The standard meter bar S is one meter long” — this is an **a priori** truth (it holds by definition), but it is **contingent** (the meter bar S could have different lengths at different temperatures; the length of one meter is a contingent physical fact)

96. Essentialism

96.1. The Rehabilitation of Essential Properties

Under the powerful critiques of Quine and others, the traditional Aristotelian distinction between “essence” and “accident” was regarded in the mid-20th century as obsolete metaphysics. Kripke’s work restored philosophical respectability to essentialism.

Kripke’s essentialism involves two key concepts:

Essentiality of Origin:

- An organism necessarily originates from the particular gametes (sperm and egg) from which it actually originated
- A table is necessarily made from the particular piece of wood from which it was actually made

Essentiality of Kind:

- If something is water, it is necessarily H_2O in every possible world
- If something is gold, it is necessarily the element with atomic number 79 in every possible world
- If something is a tiger, it necessarily has the internal biological structure of tigers in every possible world

96.2. Response to Quine

Quine had mocked essentialism: saying that a person is “essentially rational” and “contingently two-legged” depends entirely on how we describe them — mathematicians and philosophers might emphasize rationality, while acrobats might emphasize the legs.

Kripke’s response: Quine confuses **ways of referring** with **properties of the referred-to object**. We can refer to the same object in different ways, but this does not mean the properties attributed to the object are “relative to description” — objects have essential and accidental properties independently of our ways of describing them.

97. The Mind-Body Identity Theory Problem

Kripke applies his theory of rigid designators to the mind-body problem, posing a powerful challenge to mind-brain identity theory:

Mind-brain identity theory holds: pain = C-fiber firing. This is a “a posteriori identity” — like “water = H_2O .”

But Kripke points out: there is an important **asymmetry** here.

- “Water = H_2O ” — we can explain why water might **appear** not to be H_2O (because the phenomenal properties of water — transparency, drinkability — are not necessarily connected to H_2O ’s chemical structure; this is an illusion)
- “Pain = C-fiber firing” — if we feel pain, we just feel pain. Can we feel that “pain might not be C-fiber firing”? Kripke holds that this appearance is harder to explain away as illusion than the appearance that “water is not H_2O ” — because the **essence** of pain seems to be the way it is felt.

This argument shows that mind-brain identity theory faces a special challenge — at the very least, it requires more philosophical explanation than the “water = H_2O ” analogy provides.

98. Kripke’s Relation to the Western Modern Philosophical Tradition

Scholar	What Kripke Inherited	What Kripke Opposed
Frege-Russell	Retained the tradition of rigorous logical analysis of reference and meaning	Completely rejected the descriptivist view that “proper names are equivalent to descriptions” — names are not “truncated descriptions”

Quine	Inherited the tools of modal logic and quantification theory	Opposed Quine's critique of modal concepts — Quine held essentialism to be meaningless; Kripke holds that essentialism has rigorous truth conditions
Wittgenstein	Inherited the centrality of the "meaning" problem	Explicitly opposed the application of the "family resemblance" concept to natural kind terms — natural kinds possess "real essences"
Kant	Used the terminology of the "a priori / a posteriori" and "necessary / contingent" distinction	Broke the Kantian equation of "necessary = a priori" — there exist necessary a posteriori truths

99. Assigned Reading Analysis: Naming and Necessity (1972/80), pp. 22–4, 26–31, 34–6, 47–9, 71, 83–5

The assigned pages span the core passages of Kripke's three lectures, constituting his systematic critique of descriptivism and the unfolding of his own alternative theory.

99.1. Lecture 1: The Modal Critique of Descriptivism (pp. 22–4, 26–31)

99.1.1. pp. 22–4: The Modal Asymmetry Between Names and Descriptions

This is the starting point of Kripke's argument. He proposes an intuitive but devastating test: **If the meaning of "Aristotle" were equivalent to "the greatest student of Plato," then in any possible world where there is a unique person satisfying the latter, that person would be Aristotle — but this is patently false.**

Kripke points out: when we think "Aristotle might not have become a philosopher," we are thinking about **Aristotle himself** — that concrete individual who actually lived in 4th-century BCE Athens — in a scenario where he did not become a philosopher. We are not asking: "the person who satisfies 'the greatest student of Plato' might not satisfy 'the greatest student of Plato'" — that would be self-contradictory. Rather, we are asking: **this very person** might have had a different life.

From this, Kripke concludes:

- **Names are rigid designators** — they designate the same object in every possible world (as long as that object exists)
- **Descriptions are not rigid designators** — "the greatest student of Plato" could designate different people in different possible worlds

99.1.2. pp. 26–31: The Outline of the Causal-Historical Theory

In these pages, Kripke sketches his alternative theory — the **causal-historical theory of reference**. The core ideas of this theory are:

- 1) Initial "baptism": the object is assigned a name in some way (by ostension, by description)
- 2) Chain of communication: the name is transmitted from generation to generation through the communication of users
- 3) **The speaker need not possess any uniquely identifying description of the object** — as long as they stand at the end of the name's causal-historical chain, they can successfully use the name to refer to the original object

Kripke gives a vivid example: "Even if all I know about Feynman is 'he is a famous physicist' — and this also applies to many other physicists — I still use 'Feynman' to refer to Feynman himself, as long as I acquired the name from other speakers in the linguistic community, who in turn acquired it from earlier speakers... tracing back to Feynman himself."

99.2. Lecture 2: Necessary A Posteriori Truths (pp. 34–6, 47–9)

99.2.1. pp. 34–6: Breaking the Kantian Equation

The traditional view (from Kant onward, continuing to logical empiricism) holds:

- Necessary = Analytic = A Priori
- Contingent = Synthetic = A Posteriori

Kripke breaks both equations:

- “Water is H₂O” — this is an **a posteriori** truth (discovered through chemistry), but it is a **necessary** truth (in every possible world, water is necessarily H₂O — if something is not H₂O, it is not water)
- “Hesperus is Phosphorus” — this is also a necessary a posteriori truth. Because “Hesperus” and “Phosphorus” are both **rigid designators** (they both designate Venus, and in every possible world they designate the same celestial body), their identity is necessary — even though we needed astronomical observation to know this fact

99.2.2. pp. 47–9: The Essential Properties of Natural Kinds

Kripke extends the concept of rigid designator to **natural kind terms**. “Water,” “gold,” “tiger” — these terms, like proper names, are also rigid designators: they designate the same kind in every possible world where that kind exists.

Therefore:

- Gold has atomic number 79 — this is a **necessary** feature: anything that is not element 79 is **not** gold
- Tigers have a certain internal biological structure — this is also necessary: anything lacking that biological structure is not a tiger
- But the fact that gold **appears as a yellow metal** is not necessary — in another possible world, gold might appear blue due to different lighting conditions

This rehabilitates the concept of essential properties: natural kinds possess **internal essences** independent of our descriptions — essential properties are not a matter of “how we describe it,” but of “what it is.”

99.3. Lecture 3: Mind-Body Identity Theory and Rigid Designators (pp. 71, 83–5)

99.3.1. p. 71: Possible Worlds Are Not “Distant Planets”

Kripke clarifies a common misunderstanding: possible worlds are not “another world’s entities that we discover somewhere far away through a powerful telescope.” They are **ways the world could have been**, “stipulated” rather than “discovered.” This clarification clears the path for the modal arguments that follow.

99.3.2. pp. 83–5: The Mind-Body Problem

Kripke applies his theory of rigid designators and necessary a posteriori truths to mind-brain identity theory. Mind-brain identity theorists (such as J.J.C. Smart) maintain:

- “Pain = C-fiber firing” — this is an a posteriori identity, like “water = H₂O”

But Kripke identifies a crucial asymmetry: in the case of “water = H₂O,” we can explain why water might **appear** not to be H₂O — because the phenomenal properties water presents to us (transparency, drinkability) are not necessarily connected to H₂O’s chemical structure. This is an **illusion**.

But in the case of “pain = C-fiber firing,” **the essence of pain is precisely being felt in a certain way** — the appearance that “pain might not be C-fiber firing” seems harder to explain away as illusion than the appearance that “water is not H₂O.” If pain just is a certain way of being felt, then anything **felt in that way** already is pain — whether or not C-fibers are firing.

This argument does not “refute” mind-brain identity theory, but it shows: mind-brain identity theorists need to do more work than the “water = H₂O” analogy provides to explain why mind-brain identities are a posteriori but necessary.

99.4. Reading Guidance

The assigned pages constitute the core trajectory of Kripke's philosophical revolution: from the modal argument that destroys descriptivism → proposing the causal-historical alternative → breaking the Kantian equation through necessary a posteriori truths → rehabilitating essentialism → applying the results to specific philosophical problems (the mind-body problem). The logic of the three lectures advances layer by layer, each layer building on the preceding one.

100. Summary: Kripke's Philosophical Legacy

Kripke's revolutionary contributions can be summarized as:

- 1) **Names are rigid designators:** Proper names designate the same object in all possible worlds; they do not operate through descriptions
- 2) **The existence of necessary a posteriori truths:** Broke the traditional dogma that "all necessary truths are a priori" — scientific discoveries (such as "water is H₂O") reveal necessary truths
- 3) **The rehabilitation of essentialism:** Things possess essential properties independent of language (essentiality of origin, essentiality of kind); this is no longer "obsolete metaphysics"
- 4) **The causal-historical theory of reference:** Reference is established through causal chains, not through the satisfaction of descriptions in the mind of the speaker
- 5) **Possible worlds semantics:** Provided rigorous logical analysis tools for modal concepts

"Let's call something a rigid designator if in every possible world it designates the same object... When we ask whether it is necessary or contingent that *Nixon* won the election, we are asking the question whether in all possible worlds Nixon won the election." — Saul Kripke, *Naming and Necessity*